
Development of Soft Magnetic Sensors for Tri-Axial Tactile Sensing of Normal and Shear Forces and Contact Localization

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Abstract of the Thesis

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Soft tactile sensors are essential for mimicking human touch, converting physical displacements into electrical signals primarily through optical, capacitive, piezo-resistive, piezo-electric, or magnetic modalities. Specifically, magnetic tactile sensors using a magnetorheological elastomer (MRE) in which magnetized micro-particles are embedded in a soft, deformable polymer matrix provide exceptional sensitivity, durability, and a wide dynamic range for 3D force sensing. In this study, we developed a novel computational framework to map continuous magnetic-flux-density readings from an underlying Hall-effect sensor array into tri-axial signals representing normal and shear forces, and to precisely localize contact points. By accounting for the inherent non-linearities and hysteresis of the elastomer, our processing methodology ensures robust, real-time force feedback. The algorithm can be applied to MREs of varying geometric shapes and thicknesses, demonstrating its versatility across a wide range of applications. Ultimately, the sensor demonstrates accurate tactile perception, allowing the ability to detect slip, localize contact, and maintain stable manipulation of delicate objects.

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List of Abbreviations

MRE	Magnetorheological Elastomer
sMRE	Soft Magnetorheological Elastomer
hMRE	Hard Magnetorheological Elastomer
RBF	Radial Based Fhunction
PCB	Printed Circuit Board
I2C	Inter-Inetergrated Circuit
IMU	Inertial Measurement Unit
IDW	Inverse Distance Weighting

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Dedicated to my family...

Chapter 1

Introduction

Artificial touch for robotics and prosthetics has advanced significantly. However, there are still major challenges when robotic systems interact with an unstructured environment, e.g., handling a fragile object without crushing it or detecting incipient slip during rapid, dynamic grasping. In these contact-rich manipulation scenarios, accurate perception relies on sensors capable of capturing complex, multi-directional interactions rather than just rigid, single-axis pressure. Therefore, the artificial skin utilizes a compliant material specifically to achieve high sensitivity; the micro-deformations of the soft matrix under pressure or friction translate these physical forces into the high-fidelity, multi-axial feedback that the robot's control system requires. Different strategies have been employed to capture these complex interactions during manipulation to ensure safe and stable grasps. One of the strategies that might be useful to explore further is magnetic tactile sensing utilizing a magnetorheological elastomer (MRE).

MRE-based magnetic sensing differs from more traditional capacitive or piezoresistive manipulation approaches in that the transducer's electronic components are entirely decoupled from the physical contact surface. This is especially important for grasping and dexterous manipulation tasks where continuous friction and surface wear must be effectively dealt with. For example, if we were to apply a sliding force to the fingers of a robot hand, we would want to be able to detect both the downward pressure and the lateral friction simultaneously. Magnetic sensing models this

interaction through physical deformation of a smart composite material. In this way, instead of relying on rigid surface electrodes, we can treat the MR as a continuous, deformable permanent magnet. Displacing the elastomer from its resting state physically shifts these internal particles, altering the surrounding magnetic field in accordance with the applied force. This shifting magnetic flux is continuously monitored by underlying Hall-effect sensors and used to calculate precise tri-axial contact forces.

1.1 Motivation

According to a recent synthesis of the "Physical AI" technology stack, experts note that although artificial intelligence has accomplished amazing things in digital domains, such producing fluent prose and learning intricate board games, the physical world is still largely out of its grasp [1]. This disparity is an example of Moravec's Paradox: the irony that sensorimotor abilities that a human toddler can perform with ease, like grasping a delicate object or navigating uneven terrain, require far more evolutionary optimization and computational sophistication than theorem-proving or high-level strategic reasoning. [1].

This paradox is starting to change as robots move from regulated manufacturing settings to open, unstructured situations, but only if advanced multi-modal sensing and high-fidelity simulation come together [1]. Unmodeled dynamics and sensor noise that digital-only systems do not experience are frequently the source of the "reality gap"—the failure of policies trained in simulation on physical hardware [1]. Developing novel hardware and control strategies that prioritize physical agency is crucial in a world where robots must work shifts alongside people and execute intricate assembly tasks.

To bridge this gap, we propose utilizing Magneto-Rheological Elastomers (MREs) to provide the high-fidelity tactile feedback necessary for stable manipulation. By translating complex magnetic fluctuations into precise tri-axial force signals, we can

equip robotic end-effectors with the "physical intelligence" required to handle the subtle, contact-rich tasks that Moravec identified as the hardest problems in the field.

1.2 Literature Review

There are many different tactile sensors available with different modalities. Optical-based tactile sensors are one variety explored. These sensors typically rely on a camera to capture deformations of elastomers. Adelson and their group create GelSight [2], an optical based tactile sensor that used an underlying camera to track the displacement of markers on the highly deformable elastomer. Andrussov and his group created MinSight [3] that used a neural network to convert raw image input from computer vision into tactile readings .

Acoustic pressure sensors are similar in-concept to optical tactile sensors, however they are inherently different. These sensors rely on acoustic or ultrasonic waves that modulate their amplitude or time-of-flight when their propagation medium is subjected to mechanical deformation. Wettels creates an ultrasonic tactile sensor [4] using a thin compliant polymer, an acoustic medium, positioned over a piezoelectric transmitter and receiver array. Using a surface microstructure array to act as force concentrators, they can extract localized normal forces and recognize high-resolution shapes.

Capacitive-based tactile sensors are another common modality explored. These sensors rely on the change in capacitance of embedded conductive material. When a displacement is applied onto the sensor, it deforms the dielectric layer, resulting in a change in capacitance. Teyssier created a capacitive tactile sensor [5] that is made using a grid of electrodes and uses an empirical calibration to convert the change in capacitance to force readings .

Piezo-resistive tactile sensors are sensors that rely on piezo-resistive materials that change their electric resistance when they are subjected to mechanical deformation.

Sun created a piezo-resistive tactile sensor [6] that is composed of nanocomposites with pyramid shaped microstructures. This pyramid structure completes a closed circuit with its tip being the source of high resistance. When a force flattens the pyramid, the contact area of the layers increase, resulting in a drop in electrical resistance.

Piezo-electric tactile sensors are similar in-name to piezo-resistive, however they are inherently different. These sensors rely on piezo-electric materials that generate their own electrical voltage when subjected to mechanical deformation. Yu creates a piezo-electric tactile sensor [7] using polyvinylidene fluoride, a piezo-electric material, in a grid formation. Using a truncated pyramid bump to improve force transmission, they can extract normal and shear forces.

Magnetic-based tactile sensors are mainly based on mechanical deformation that alters a magnetic field. A magnetic sensor underneath is used to record the magnetic field. This sensor is typically a hall-effect sensor or a 3D magnetometer. There are two main magnetic-based tactile sensor variations, denoted based on their magnetic field source. One of these variations relies on tiny embedded permanent magnets. When a mechanical deformation is applied to the elastomer with the embedded permanent magnets, these magnets shift in predictable ways and can be converted to readings.

U-skin, created by Tomo and their team, uses an array of magnetometers with embedded permanent magnets right above each magnetometer. In their first iteration with a flat silicone [8], they found that the incompressibility of the silicone was causing unwanted readings along the x-y axes. When a normal force is applied not directly above the magnet, the magnet would shift in the x-y plane, giving a sense of shear even when its not present. In their second iteration where they incorporated air gaps [9], they were able to reduce this affect with their unique silicone design.

Not all soft magnetic sensors use silicone as their medium. E-flesh [10], instead uses 3D-printed compliant mechanisms to simulate the elastic effect. To obtain a magnetic field for sensing capabilities, they embedded permanent magnets during the printing process. By tuning the parameters of the repeating unit cells, primarily the thickness

and size of each cell, they can achieve desired mechanical properties. By utilizing machine learning techniques, they were able to achieve sensing capabilities.

The other variation of magnetic sensors, the one our sensor uses, utilizes MREs. An MRE is a smart composite material that consists of a highly compliant rubber-like polymer matrix embedded with magnetized micro-particles [11]. MREs sometimes utilize materials that don't retain their magnetic properties, such as iron. These versions of MREs are called soft-MREs (sMRE) [12]. These are typically used to exhibit different mechanical properties depending on the magnetic field and are common in vibration applications. MREs can also use particles that do retain their magnetic field, such as Neodymium. These MREs are called hard-MREs (hMRE) [13]. These particles have high remanence and coercivity, meaning they have high residual magnetism and have high resistance to being un-magnetized. MREs can also be classified into two different groups, isotropic and anisotropic. For an isotropic MRE, the micro-particles are uniformly mixed throughout the MRE. As a result, this MRE exhibits homogeneous physical behaviors in every direction. An anisotropic MRE instead aligns their magnetic particles during the manufacturing process. This results in a MRE that is more responsive along the polarization. [14].

An example of an anisotropic hMRE sensor is Bhirangi's ReSkin [15] sensor. Their hMRE was made by mixing NdFeB melt-spun powder with silicone in a 1:1 ratio and magnetized during the curing process. They used a grid of magnets instead of a stronger permanent magnet to take advantage of strong-edge effects of permanent magnets. To counteract the unique magnetic field the resulting hMRE would have, they relied on machine learning to obtain force and localization readings. Bhirangi also experimented with an isotropic hMRE, AnySkin [16]. In this version, they used a pulse magnetizer to magnetize the skin post-curing. This provided them a more uniform magnetic field for their sensor. This sensor also utilized machine learning to obtain their results.

Our design presents an alternative method in converting magnetic readings into

force and localization estimations. We maintained a similar hMRE to Bhirangi's Re-Skin design while taking some improvements he made in his AnySkin design while also providing some alternative improvements of our own. One main limiting factor in the hMRE sensor designs provided by Bhirangi is the inability to detect multiple point contacts. Our sensor is capable of doing this ability.

1.3 Contributions

The contributions of our soft magnetic sensor are as follows:

- Development of a cost-effective manufacturing process for hard magnetorheological elastomers (hMREs) utilizing melt-spun NdFeB powder and permanent magnets to produce a strong magnetic field.
- Implementation of a spatial mapping approach using a Gaussian Radial Basis Function (RBF) to model magnetic displacement topology for the accurate detection and localization of multiple simultaneous contact points.
- Formulation of a computational framework to decouple and estimate tri-axial forces by tracking the spatial shift of the magnetic centroid for the detection of direction/magnitude of shear forces.
- Integration of a dynamic baseline subtraction methodology to continuously filter out ambient magnetic interference and ensure reliable tactile readings in unstructured environments.

1.4 Thesis Outline

The remainder of the thesis will be organized as follows: Chapter 2 will go over the magnetic models that are used in our sensor. Chapter 3 will go over how our sensor was created, including the MRE, electrical, and software components. Chapter 4 will

go over the experimental results gathered from our sensor. Lastly, Chapter 5 will be a final discussion and plans for future works.

Chapter 2

Theoretical Background

2.1 Relative Data Approach

In theoretical magneto-static models, we typically assume that ambient magnetic fields are not present. However, when dealing with real-world applications, this is not the case. There are large magnetic fields that can alter readings, despite being far away, the Earth's magnetic field being a prime example.

Lets assume we have a constant noise applied to our magnetic field readings, B_n . If we were to capture a baseline reading, this reading would capture our wanted magnetic reading, B_0 and a constant noise as modeled below:

$$B_{0,n} = B_0 + B_n \quad (2.1)$$

Where $B_{0,n}$ is the noisy baseline reading. Based on the assumption that the magnetic field noise is constant, we can also assume any future readings $B_{t,n}$, will include the same noise, B_n , modeled as:

$$B_{t,n} = B_t + B_n \quad (2.2)$$

Where B_t is the magnetic field we wanted to capture after some time t . By taking the difference between our noisy reading and our noisy baseline, we get the following equation:

$$\Delta B = B_{t,n} - B_{0,n} \quad (2.3)$$

Where ΔB is the change in the magnetic field. By substituting equations 2.1 and 2.2 into equation 2.3, we can get the following equation:

$$\Delta B = (B_t + B_n) - (B_0 + B_n) \quad (2.4)$$

$$\Delta B = B_t - B_0 \quad (2.5)$$

Notice how in equation 2.5, the noisy magnetic field, B_n is no longer present. By using this new parameter that is still dependent on the magnetic field of the MRE, we can get cleaner magnetic readings for our sensor.

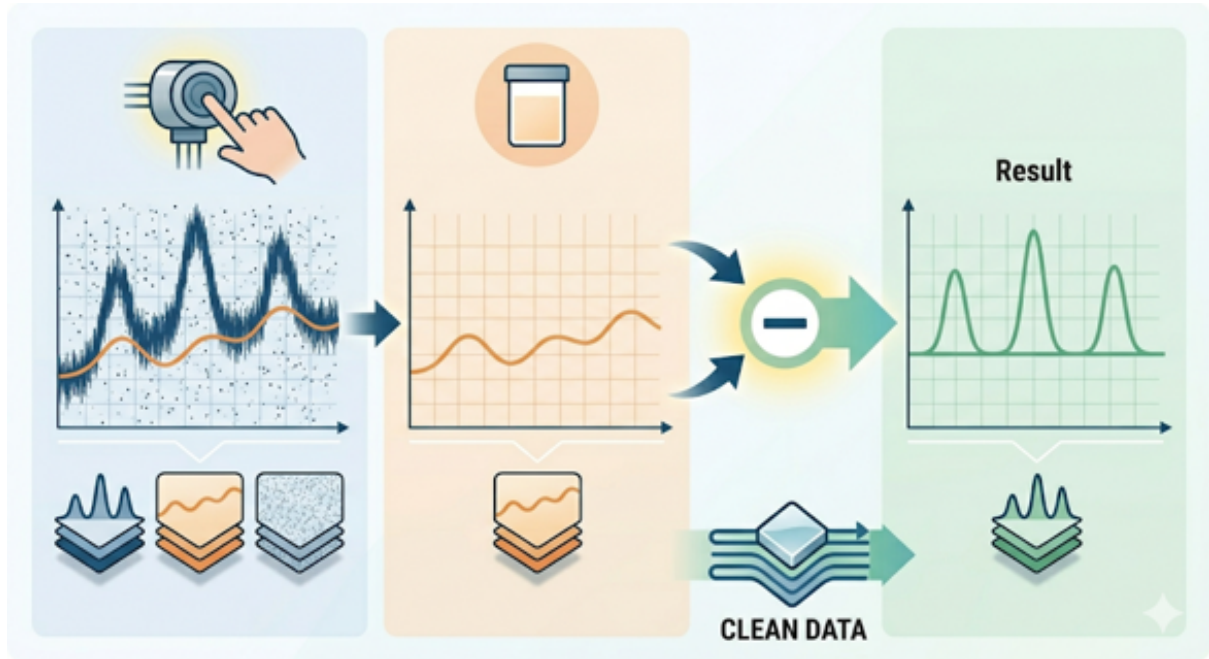


FIGURE 2.1: Relative Data Approach

2.2 Point Localization

2.2.1 Weighted Centroid

To effectively utilize a magnetometer array, it is necessary to accurately translate the raw sensor data into a localized spatial coordinate. While various complex algorithms exist for tactile localization, a computationally efficient and deterministic approach is often preferred over heavier machine learning models. Therefore, a method to obtain the point contact on our sensor is using a weighted centroid method. Ciobanu [17] examined the possibilities of using weighted averages to calculate a point contact for the BioTac sensor, instead of using a neural-network as was used on the original paper [4]. The approach, modified to fit our sensor, relied on the following formulas to obtain a point localization, $p_c = (x_{pc}, y_{pc}, z_{pc})$:

$$x_{pc} = \frac{\sum_{i=1}^k (w_i x_i)}{\sum_{i=1}^k w_i} \quad (2.6)$$

$$y_{pc} = \frac{\sum_{i=1}^k (w_i y_i)}{\sum_{i=1}^k w_i} \quad (2.7)$$

$$z_{pc} = \frac{\sum_{i=1}^k (w_i z_i)}{\sum_{i=1}^k w_i} \quad (2.8)$$

Where x_i , y_i , and z_i is the coordinate for each of the k amount of magnetometers. Due to our sensor being flat along the x-y plane, our z_i will always be 0, thus z_{pc} will as well. For our weights, w_i , is calculated as follows:

$$w_i = \|\Delta B_i\|_2 \quad (2.9)$$

In physical terms, this algorithm treats the array of magnetometers much like a center-of-mass calculation, where the 'mass' at any given point is the magnitude of the magnetic field disruption, $\|\Delta B_i\|_2$. When a force is applied to the sensor's surface, it causes a localized displacement within the hard magnetorheological elastomer

(MRE), pushing the embedded magnetic particles closer to the underlying electronics. Because these magnetic field disruptions decay rapidly with spatial distance, the magnetometer situated most directly beneath the contact point experiences the most drastic shift in its localized field, while sensors further away register minimally. By utilizing this physical decay and weighting the data with $\|\Delta B_i\|_2$, the algorithm naturally biases the spatial 'balance point' toward the strongest readings. This allows the system to smoothly interpolate the exact point of maximum physical compression, pulling the final x_{pc}, y_{pc} coordinate precisely toward the center of the deformation.

2.2.2 Radial Basis Function

While a weighted centroid method provides a computationally lightweight approximation for a single contact point, it is fundamentally limited by its inability to resolve simultaneous deformations. To achieve high resolution and topological accuracy, we propose a spatial mapping approach utilizing a Gaussian Radial Basis Function (RBF) applied over a discretized virtual grid.

For our sensor, the hMRE utilized functions as a deformable permanent magnet. When mechanical force is applied to the surface, the elastomer deforms, locally displacing the permanently magnetized microparticles.

To model this interaction, the displaced region can be approximated as a magnetic dipole. In classical physics, the magnitude of the magnetic field, B , generated by a dipole at a distance r along its central axis is defined as:

$$B = \frac{\mu_0}{4\pi} \frac{2m}{r^3} \quad (2.10)$$

Where μ_0 is the magnetic permeability of free space, m is the magnetic dipole moment, and r is the spatial distance from the center of the dipole to the observation point. Unlike a theoretical monopole, a dipole consists of a North and South pole separated by a microscopic distance. As the observation distance r increases, the opposing

forces of the two poles rapidly cancel each other out, resulting in a field strength that decays proportionally to the inverse cube of the distance ($B \propto 1/r^3$).

Consequently, when the MRE is indented, even a microscopic reduction in distance r produces a massive, highly localized spike in the measured magnetic field. Across the discrete 2D sensor array, this inverse-cube decay creates a radially symmetric, sharp activation footprint. While the strict physical decay follows the $1/r^3$ law, a Gaussian mathematical kernel serves as an optimal empirical approximation for this footprint. It provides a continuous, non-linear interpolation that effectively models the sharp spatial boundary of the physical deformation, allowing us to accurately reconstruct the topological profile from discrete sensor readings.

To determine the precise localization point, we define a high-resolution 2D coordinate grid across the sensor's surface. For every probed position $p = (x, y)$ on this virtual grid, we compute an activation value, $V(p)$, which represents the estimated localized magnetic displacement at that coordinate. This value is calculated by aggregating the Gaussian RBF contributions from all k magnetometers. The amplitude of each sensor's Gaussian curve is driven by the magnitude of the measured local magnetic field change, shown in equation 2.9. The activation value at any grid point p is defined as:

$$V(p) = \sum_{i=1}^k w_i \cdot \exp\left(-\frac{\|p - c_i\|^2}{2\sigma^2}\right) \quad (2.11)$$

Where p is the coordinate of the probed position of the grid, c_i is the physical coordinate of the i -th magnetometer, and $\|p - c_i\|^2$ is the squared Euclidean distance between the grid point and the sensor. σ^2 is the variance parameter which must be empirically tuned to represent the spatial width of the deformation footprint, matching the physical spread of the MRE's localized displacement.

To extract a topological shape of the deformation from the applied force, the computed grid values $V(p)$ are subsequently normalized. We scale the entire grid to a

standardized range of [0,1] using min-max normalization:

$$V_{norm}(p) = \frac{V(p) - V_{min}}{V_{max} - V_{min}} \quad (2.12)$$

Where V_{min} and V_{max} are the minimum and maximum activation values found across the entire computed grid. This step converts the raw magnetic variance data into a localized intensity map.

Once the normalized surface map $V_{norm}(p)$ is generated, the spatial localization is achieved by evaluating the grid to find the local maxima. Because the RBF algorithm mathematically reconstructs the topological "hills" of the localized magnetic displacement, a local maximum directly corresponds to the epicenter of the applied force. By extracting the (x, y) coordinates of these peaks, the algorithm accurately pinpoints the exact point of contact. Furthermore, evaluating the data across this continuous RBF grid inherently allows for the detection of multiple distinct local maxima, establishing a foundation for multi-touch localization.

2.3 Shear Force Measurement

While normal force localization relies on the localized variation in the magnetic field ΔB_i to map structural indentation, estimating shear force requires tracking the lateral translation of the elastomer. To achieve this, we utilize a displacement vector derived from the spatial shift of the sensor's absolute magnetic centroid, utilizing a baseline centroid method.

When a lateral shear force is applied across the x-y plane of the sensor, the hMRE undergoes elastic shear strain. This physical translation drags the embedded NdFeB permanent magnetic material laterally relative to the underlying magnetometer array. The mechanics of this lateral deformation are governed by Hooke's Law for shear,

which states that the applied shear stress τ is directly proportional to the resulting shear strain γ :

$$\tau = G\gamma \quad (2.13)$$

Where G represents the shear modulus of the hMRE, a constant dictating the material's specific resistance to shearing deformation. The shear strain γ is mathematically defined as the lateral displacement distance Δx divided by the uniform thickness or height of the elastomer h . Substituting this definition into the continuous material formulation of Hooke's Law yields:

$$\tau = G\left(\frac{\Delta x}{h}\right) \quad (2.14)$$

Rearranging this equation to isolate the physical displacement Δx demonstrates the core operating principle of our shear detection:

$$\Delta x = \frac{\tau \cdot h}{G} \quad (2.15)$$

Because both the thickness, h , and the shear modulus, G , of the elastomer are fixed physical constants, the lateral spatial displacement Δx is strictly linearly proportional to the applied shear stress τ . Therefore, by mathematically tracking the continuous lateral shift of the MRE's absolute magnetic field, we can extract a direct, proportional measurement of the applied shear force.

For normal force localization, evaluating the differential change in the magnetic field ΔB_i is optimal because it isolates the localized indentation of the MRE, ignoring the baseline field. However, applying ΔB_i to lateral shear introduces a complex, bipolar artifact: as the magnetic material translates laterally, sensors on the leading edge and trailing edge register magnetic readings. This creates a noisy, asymmetrical profile that obfuscates the true physical translation.

By instead calculating the centroid using the absolute magnitude of the raw magnetic field B_i , the algorithm effectively tracks the global "center of mass" of the permanent magnet. As the elastomer shears, this bulk magnetic center translates laterally in exact unison with the physical material, providing a clean, stable vector that accurately reflects the physical displacement described by Hooke's Law.

To isolate this lateral shear vector, we calculate a 2D weighted centroid using the absolute field magnitude. This relies on the relative data approach to establish a zero-state reference point for the lateral displacement. However, instead of using the magnetic fields, we will instead be setting a baseline for the centroid.

First, a baseline centroid, $p_{c,base} = (x_{base}, y_{base})$, is established and stored in memory during an unloaded calibration phase. The weights for this baseline state are defined by the absolute magnitude of each magnetometer's reading at rest, $w_{base,i} = ||B_{base,i}||_2$:

$$x_{base} = \frac{\sum_{i=1}^k (w_{base,i} x_i)}{\sum_{i=1}^k w_{base,i}} \quad (2.16)$$

$$y_{base} = \frac{\sum_{i=1}^k (w_{base,i} y_i)}{\sum_{i=1}^k w_{base,i}} \quad (2.17)$$

During operation, an active centroid, $p_{c,act} = (x_{act}, y_{act})$, is continuously computed using the absolute magnitude of the real-time magnetic field readings, $w_{act,i} = ||B_{act,i}||_2$, applied to the centroid formulas 2.16 and 2.17.

The physical lateral deformation of the MRE is extracted by calculating the 2D displacement vector, $\vec{d}$, between the active centroid and the stored baseline centroid:

$$\vec{d} = p_{c,act} - p_{c,base} \quad (2.18)$$

This dynamic displacement vector $d = (\Delta x, \Delta y)$ directly represents the physical shift derived in Hooke's Law. To compute the final applied shear force F_{shear} , we expand the shear stress definition, 2.19, where A is the contact area. Substituting this into our Hooke's Law derivation proves that the applied force is linearly proportional

to the displacement magnitude:

$$\tau = F_{shear}/A \quad (2.19)$$

$$F_{shear} = (\frac{A \cdot G}{h}) ||\vec{d}|| \quad (2.20)$$

$$\theta = \arctan2(\Delta y, \Delta x) \quad (2.21)$$

By continuously computing the absolute-field centroid displacement vector $\vec{d}$, the system simultaneously extracts the directional angle of the lateral force shown in 2.21 and a linear scalar magnitude shown in 2.20. This methodology effectively decouples shear translation from normal compressive deformation, enabling continuous, robust multi-axis force estimation.

2.4 Normal Force Measurement

To accurately estimate the magnitude of the applied normal force, the system calculates the discrete spatial integral of the localized magnetic displacement. This is achieved by summing the absolute variation of the vertical magnetic field component across the entire sensor array.

Our sensor architecture features a hMRE positioned directly flush against a rigid PCB housing the magnetometer array. Because the bottom boundary is fixed, an applied normal force does not cause the bulk material to translate vertically; rather, it induces a complex strain gradient within the silicone matrix.

Due to the fundamental incompressibility of elastomers, the volume of the MRE must remain constant under load. Therefore, when a normal force compresses the upper layers of the elastomer downward, the internal material is physically forced to displace laterally, bulging outwards radially from the epicenter of the applied force.

This internal mechanical deformation causes a spatial rearrangement of the embedded NdFeB magnetic dipoles—the upper dipoles are slightly rotated and pushed closer to the PCB, while the central dipoles are swept laterally outwards. This combined vertical compression and radial expansion creates a complex, multi-directional shift in the magnetic flux density measured at the surface of the sensor array. As the applied normal force increases, both the depth of the vertical compression and the radius of the lateral bulging increase proportionally.

To mathematically capture this complete internal volumetric deformation, we isolate the Z-axis component of the magnetic field, B_z . We calculate the differential change, $\Delta B_{z,i}$, for each i -th sensor by subtracting its resting baseline reading from its active state. Because the internal bulging of the MRE pushes magnetic material both toward certain sensors and away from others, the localized Z-axis vector can shift positively or negatively. If these raw values were simply summed, the positive and negative variations caused by the radial bulging would destructively interfere, artificially diminishing the measured force. To ensure that all internal spatial displacement contributes constructively to the total measurement, we extract the absolute magnitude of the change at each coordinate. A scalar representation of the total volumetric magnetic displacement, S_{mag} , is calculated via summation:

$$S_{mag} = \sum_{i=1}^k |\Delta B_{z,i}| \quad (2.22)$$

While S_{mag} captures the total magnetic variance, its relationship to the applied normal force F_{norm} is inherently non-linear. This non-linearity is driven by two primary factors: the hyperelastic stiffening of the silicone matrix at higher strains, and the exponential $1/r^3$ near-field behavior of the magnetic dipoles as they approach the magnetometer array. To accurately resolve the normal force, the system maps the magnetic summation S_{mag} through an empirically derived non-linear calibration function, $f(S_{mag})$:

$$F_{norm} = f(S_{mag}) \quad (2.23)$$

Where $f(S_{mag})$ is a polynomial regression curve extracted from controlled physical load-testing. This non-linear mapping effectively translates the aggregated magnetic flux variations—accounting for both geometric bulging and exponential field interactions—into a highly accurate, calibrated measurement of normal force in Newtons.

Chapter 3

Sensor Manufacturing

3.1 Sensor Iterations

In one of our first iterations, we initially experimented with an embedded magnet variation. Similar to how Tomo made their prototype [8], we experimented with something similar. We iterated by including the bump method Yu and their team experimented with [7], and used a larger permanent magnet, 0.25". In this iteration, we noticed that the change in magnetic fields along the z-axis was not as sensitive as we hoped. We believe that the displacement of the large magnet was not as noticeable compared to a smaller magnet. In addition, a drawback of the bump was that localization was very difficult due to the bump spread out our force.

In our second iteration, we add an air gap, similar to Tomo's second iteration [9] while maintaining the single magnet and bump. In this iteration, we noticed that despite getting larger z-axis deformations, shear force estimation became difficult. This was due to the new air gap, making it such that the magnet may laterally shift, but it may also rotate. This coupling issue proved difficult.



FIGURE 3.2: Prototype hMRE

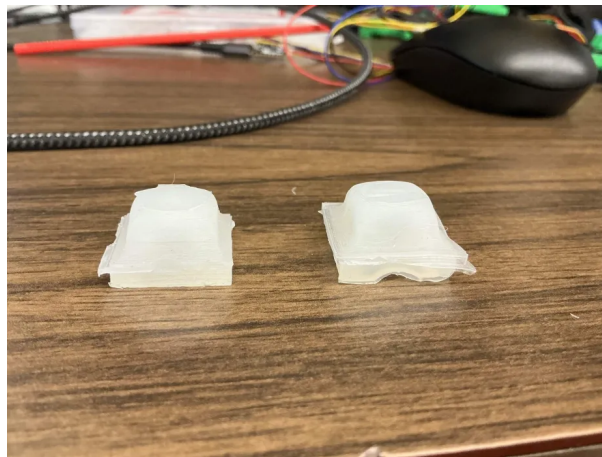


FIGURE 3.1: Old Sensor Iterations

Lastly, we decided to pivot entirely the hMRE we ended up on. We originally manufactured the same hMRE Bhriangi and his team made for their ReSkin iteration [15]. However, the magnetic field of the hMRE became very chaotic due to their magnetization method. In their design, they claimed to use a grid of 0.25" cube magnets. In order to get the grid formation they showed, it required the cubes to be formed into an alternating pattern. This alternating pattern is had high variance, as stated in their next iteration, [16]. So we decided to alter the magnetization method, without relying on the expensive pulse magnetizer they mentioned in their next iteration.



FIGURE 3.3: Silicone and NdFeB powder used

3.2 Manufacturing MRE

Our final hMRE utilizes a 25 μ m melt spun NdFeB powder and DragonSkin 10 Slow silicone, same as our older iteration but changes the magnetization method to provide a stronger, more uniform magnetic field, and a thicker skin for more deformation to occur.

There are two manufacturing methods for Neodymium Iron Boron powder, either sintering or melt-spun. Sintered NdFeB is manufactured by milling the alloy into a fine powder where each particle acts as a single magnetic domain. These particles are poured into a mold, aligned using a massive external electromagnet, and then pressed and baked (sintered) into a fully dense, solid block. To get the powdered NdFeB, the manufacturer have to crush that finished block. This physically shatters the carefully baked microscopic grain boundaries, fundamentally destroying the internal magnetic structure.

However, melt-spun powder is manufactured by shooting directly unto a rapidly spinning, hyper-cooled copper wheel. The liquid flash-freezes in milliseconds into a thin, brittle ribbon. This extreme rapid freezing forces the alloy to crystallize into millions of microscopic, randomly oriented nanograins within the ribbon itself. The ribbon is then easily pulverized into the powder. When the melt-spun ribbons are

crushed into powder, the fractures happen around the microscopic nanograins, not through them. This guarantees that the raw powder brings exceptional coercivity to the table and will act as a strong, true permanent magnet.

According to Bhriangi [16], he noted that with this specific combination of silicone and particle size, their elastomer had enough time to cure before the particles could settle on one surface of the elastomer. However, this is only valid when the hMRE being made is isotropic. In this research we plan to experiment with both anisotropic and isotropic mediums. Despite the pivot, we still plan to use the findings from Bhrinagi and his team to maintain consistence between the anisotropic and isotropic hMREs.

To enable high sensitivity for our sensor, a low shore hardness is ideal. The shore hardness is a standardized measurement that determines a material's resistance to indentation. It is the most common way to measure the stiffness, flexibility, and physical softness of rubbers, elastomers, and polymers. With a light force, the deformation of the silicone will be much greater compared to a higher shore hardness material. To provide enough time to manufacture, vacuum, and pouring, a very slow variation would guarantee enough time to get the hMRE mixture into its mold.

Although vacuuming is the best way to remove air bubbles in the mixture post mixing, there is another technique that can help remove as many air bubbles as possible. This technique involves pouring the mixture from a high place and letting a very thin stream pour onto the mold. This thin stream basically filters any air bubbles that may still be present in the mixture. Despite this technique, vacuuming still provided the best solution to removing bubbles in the mixture.

For the magnetization method, instead of using a grid like pattern for our magnetization method, we instead decided to use strong single magnets. For our magnetization, we decided to use 5/4" diameter, 1/4" thick N52 magnets 3.4. This allowed us to maintain relatively uniform magnetic fields. According to KJ Magnetics' extensive product testing, a magnet of this size and grade has a surface field strength of 2.748 gauss or 0.2748T [18]. However, due to the increase in grade of the magnets, the pull

force of these magnets is much stronger, making it much more difficult to pull apart, at 110.7 lbf according to KJ Magnetics [18].

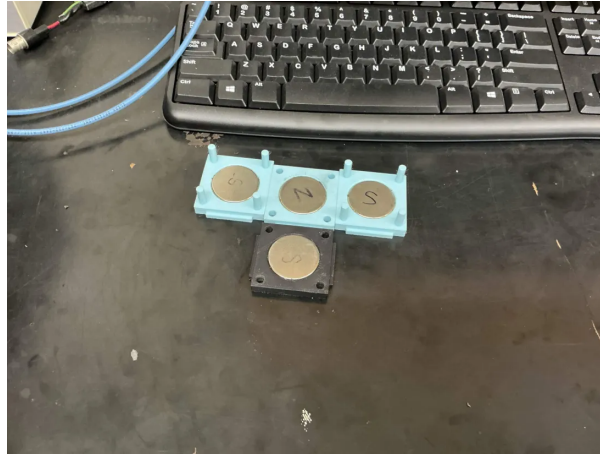


FIGURE 3.4: N52 Magnets with their respective holders

Despite the strong magnetic field of the N52 magnets, the remembrance of the Nd-FeB powder is around 1.19T based on Magnequench's similar product [19]. Even when sandwiching the hMRE, the total hMRE will experience only 0.5496T. Due to this difference, the hMRE we will be using will not be at its full possible strength.

The first step is to spray mold release onto our mold. This allows us to easily pull off the hMRE when its finishing curing. The mold release used was AG Adhesive Guru's mold release.

To make our mixture, the DragonSkin 10A silicone comes in parts A and B. We mix the parts A, B, and the NdFeB particles in a 1:1:2 ratio. For an easy mixing operation, we first mixed either Part A or B with the NdFeB powder. After thoroughly mixing, we mixed in the remaining part.

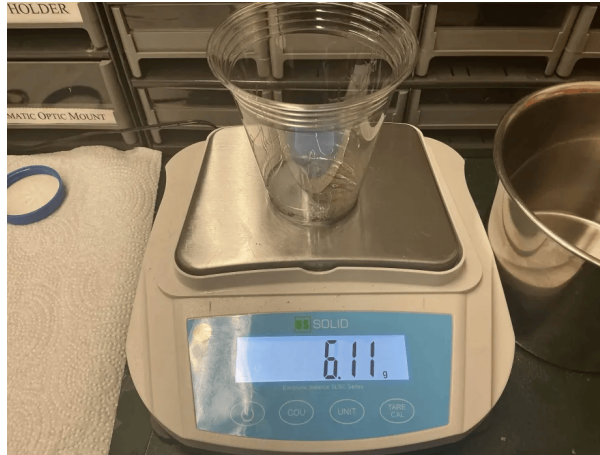


FIGURE 3.5: Mixing the Silicone and NdFeB powder

The next step is to pull a vacuum on our silicone mixture. The vacuum we used was BACOENG's 2 quart vacuum chamber. We pulled the vacuum for roughly 5 minutes or until the mixture has stopped bubbling.



FIGURE 3.6: Caption

After pulling the mixture out of the vacuum chamber, to get the silicone into the mold we used the pouring technique described previously. By pouring at least a foot above the mold, we can achieve the thin stream that will filter any air bubbles. To get any final bubbles, we tap the mold slightly.

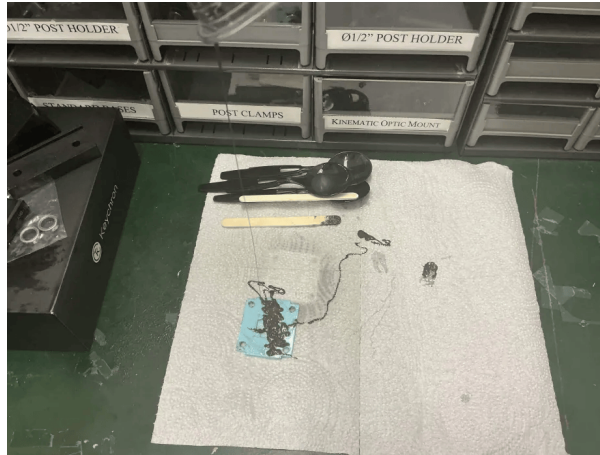


FIGURE 3.7: Pouring Technique

After pouring, a thin plastic film was laid on top of the silicone. This was done to achieve a nice surface finish. Depending on whether or not we want an anisotropic or isotropic hMRE, we either sandwich this mold between our two N52 magnets or wait after its finished curing and then sandwich it.



FIGURE 3.8: Magnetizing via N52 Magnets

After the hMRE has finished magnetizing and curing, the next step is to pull it out of the magnetization apparatus. To remove the magnets, we used the help of an optical table to remove them. Essentially, we used the table to attract the magnets to be able to shear them off the hMRE and each other.



FIGURE 3.9: Pulling MRE out of N52 Magnets

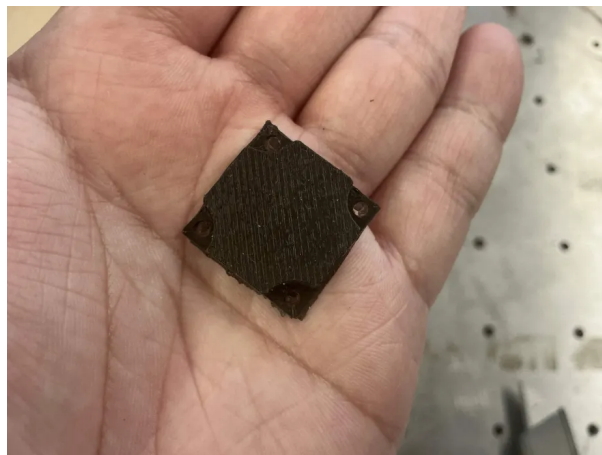


FIGURE 3.10: Manufactured MRE

3.3 Electrical Hardware

For our PCB design, we decided to adapt Bhriangi's PCB design from his ReSkin [15], and AnySkin [16] sensors. Their PCB design utilizes 5 MLX90393 magnetometers, where 4 of them are 7 mm apart in the cardinal directions from a central magnetometer, shown in figure 3.11. The MLX90393 magnetometers are able to transmit stable 3-axis magnetic flux density readings at around 500 Hz.

To be able to obtain information from multiple PCBs at the same time, we decided to use a PCA9548A Multiplexer. This multiplexer allows us to communicate with multiple copies of the same PCB that repeat the I2C address of each magnetometer.

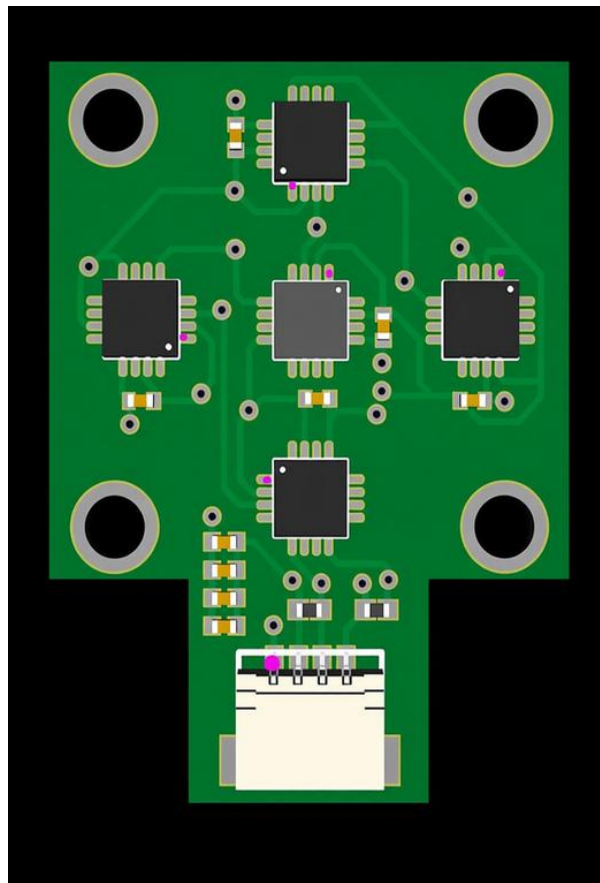


FIGURE 3.11: 2D Render of PCB by Bhriangi [15]

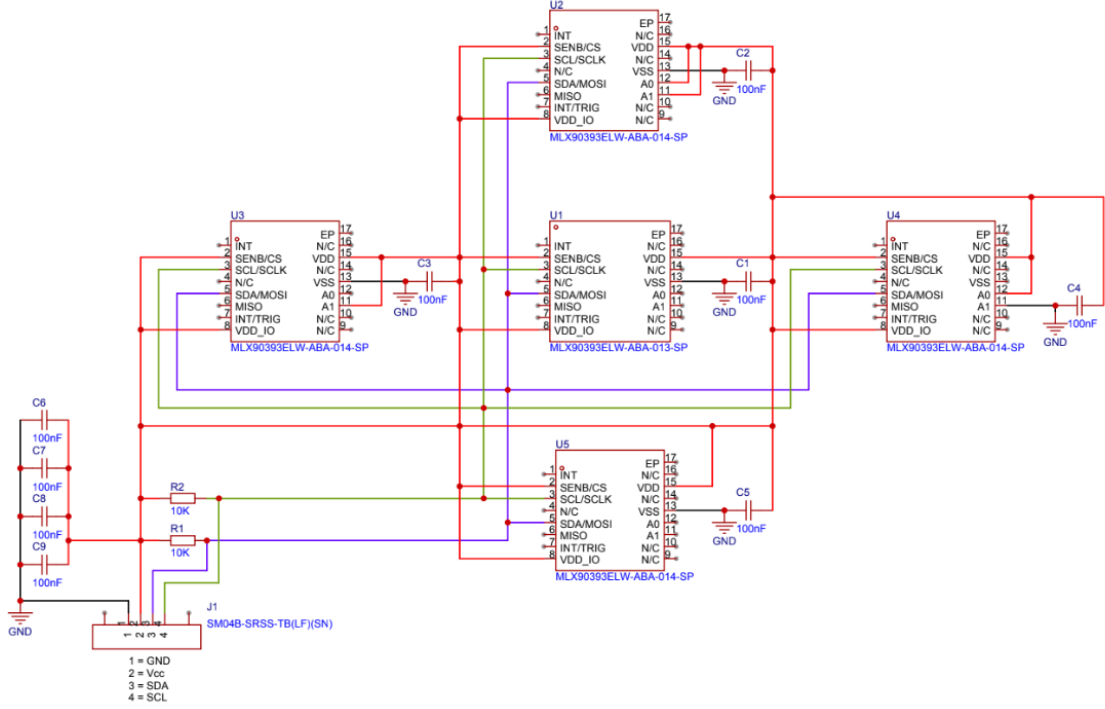


FIGURE 3.12: Single PCB Schematic by Bhrinagi [15]

3.4 Control Software

The code begins by continuously reading structured serial frames from the Arduino at 2 million baud. Each frame contains raw IMU readings from up to five sub-sensors per channel, and only the most recent complete frame in any given read burst is kept — ensuring the pipeline always works with the freshest data available.

Those raw readings immediately undergo rotation correction, since each sub-sensor is physically mounted at a different angle. A mirrored rotation remaps the axes of each sensor into a common canonical frame, followed by two sign-flip passes to account for face-down PCB orientations. This corrected vector is what enters the filter chain.

The corrected readings then pass through a two-stage filter. First, a median filter holds the last 5 samples in a sliding window and takes the element-wise median, killing any impulse spikes or vibration artifacts. That median feeds directly into an exponential low-pass filter $\alpha = 0.2$ applied to the deviation from the zeroed baseline,

producing a smoothed differential vector (d_x, d_y, d_z) that only reflects sustained contact rather than transient noise.

From that smoothed differential, each sensor is reduced to their respective weights, after any component below the user-set dead-band threshold is zeroed out. Sensors with negligible weight are dropped entirely before the next stage.

The contact localization is then computed directly from those weights as a weighted centroid across all active sensors. Each sensor's world-space position is configured by the user in the position table and the result is a single world-coordinate point that tracks the weighted center of all contact activity across the entire skin surface simultaneously.

Alternatively, those weights and their physical canvas positions may also be used to build a continuous pressure field. The gaussian kernel is evaluated for each active sensor across a 60×60 grid of canvas coordinates. All sensor kernels are summed together and normalized, producing a smooth scalar field where high values indicate regions of heavy contact.

That field is then scanned for contact points. Any cell in the grid whose normalized value exceeds 0.15 and is the strict maximum within its 3×3 neighborhood is flagged as a candidate peak. Candidates are sorted by strength and a greedy merge pass drops any peak that falls within 3 cells of a stronger one, preventing a single contact blob from producing multiple spurious markers. Each surviving peak is then refined to sub-cell precision by computing a weighted centroid over its surrounding 3×3 patch - using the field values themselves as weights - yielding a fractional grid index finer than any single cell.

Another alternative is to compute the weighted centroid of the raw sensor readings, post-rotation and 2-stage filter. The centroid calculations are exponentially smoothed each frame via an exponential moving average, reducing jitter without significant lag. When the user zeros the sensors, this smoothed centroid is frozen as a baseline. Every subsequent frame, the shear displacement is computed as the 2D vector that describes

the intensity and angle of the applied shear. If the magnitude exceeds a configurable saturation limit, the vector is clamped to that radius and flagged visually.

The $\sum |B_z|$ value is computed by taking the smoothed differential z component from every active sensor across all channels, applying the absolute value to each, and summing them into a single scalar value.

We also incorporated an auto-zeroing system to automatically re-calibrate the baseline whenever the skin has been at rest long enough. Every poll tick it computes the rate of change of each sensor's diff vector as a proxy for whether the sensor is still moving. If that derivative falls below the derivative threshold and the overall reading magnitude is below the sensor's individual cutoff, the sensor is considered settled. Once it has remained settled continuously for the configured settling time the zeroing is called automatically, snapping the current reading as the new baseline.

Our final code is located in the Github in the appendix A.2.

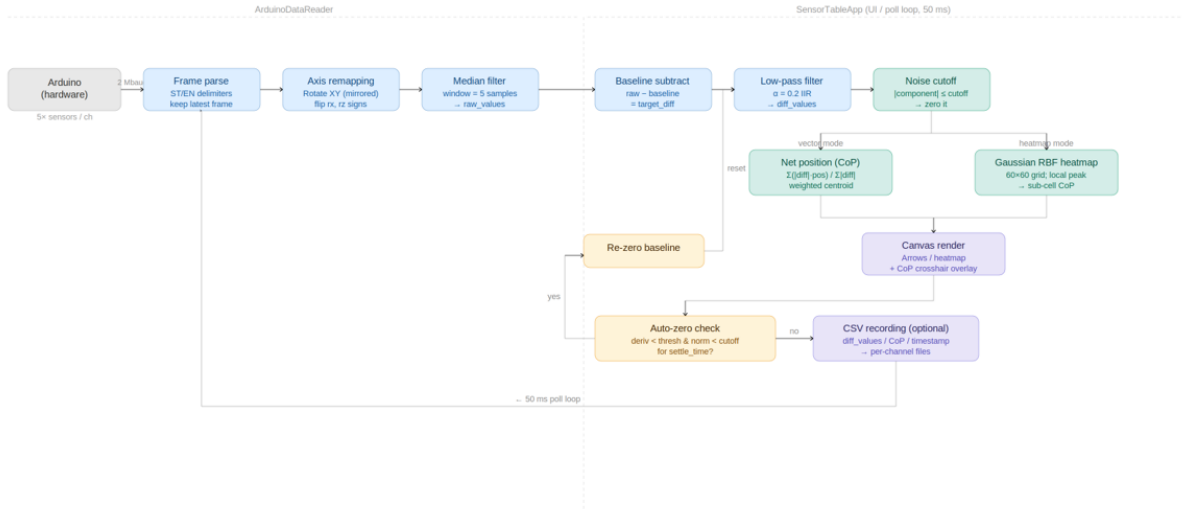


FIGURE 3.13: Software Flowchart

3.5 Localization MRE Design

The MRE we used in our localization assessment, is 40mm by 40mm by 3mm. For this specific design, due to the size of our magnets, we must use a isotropic variant,

meaning we had to wait for the hMRE to finishing curing. The open-air mold is designed such that the smooth surface finish of the elastomer is against the PCB. We also included holes in the corners of the hMRE so that it can be easily connected to the PCB via an M2 screw. The mold also included a flat area for the magnets to sit while we take the hMRE out.

After following the aforementioned steps and getting the hMRE, we must magnetize. Ideally, the N52 magnets would be large enough such that it covers the entire elastomer to get as even of a magnetic field distribution as possible, however, we only had access to the N52 magnets mentioned previously. To magnetize the entire region of the hMRE, we moved the magnets along the hMRE in a uniform pattern. After finishing the pattern, we shifted the magnets away from the hMRE and extracted the magnetized hMRE.

For our localization hMRE, we used 4 of the PCBs, mounted in a square like fashion. To hold them steady, screws and guide extrusions were used.



FIGURE 3.14: MRE used for Localization Tests

3.6 Shear MRE Design

The MRE we used in our shear assessment is more unique. Instead of our entire sensor being an MRE, only the part making contact is. In between the hMRE and PCB is

normal silicone, the same one used in the process hMRE making process. The hMRE section is 20mm by 20mm by 3mm, anisotropic, and is pre-magnetized before further manufacturing. After making this hMRE using the same process mentioned before, a normal silicone is made with the dimensions, 20mm by 20mm by 5mm. This silicone is made by following the same process besides adding the NdFeB powder. To attach the two elastomers together, a silicone adhesive, Smooth-On's SIL-Poxy, was used to merge them together.

The reason for adding the normal silicone section was to dramatically increase the shift of the magnetic field. If the hMRE is directly against the PCB, the lateral shift is much lower.



FIGURE 3.15: Shear MRE Design

3.7 Final Design Summary

In conclusion, the fabrication of the hMRE sensor relies on a precise integration of low-shore-hardness DragonSkin 10 Slow silicone and 25 micron melt-spun NdFeB powder, combined in a 1:1:2 ratio. Refer to the bill of materials in appendix A.1 for more information on them. To ensure a uniform, high-quality elastomer matrix without structural defects, the mixture undergoes rigorous degassing via a vacuum chamber combined with a specialized high, thin-stream pouring technique to eliminate air bubbles.

Depending on the desired structural alignment-isotropic or anisotropic-the elastomer is cast into a prepared mold and magnetized using powerful N52 neodymium magnets either after or during the curing process. Finally, the completed hMRE is physically integrated with the magnetometer-equipped PCB array—sometimes utilizing an intermediate layer of pure silicone to amplify lateral magnetic shifts-resulting in a highly responsive, customized tactile sensor.

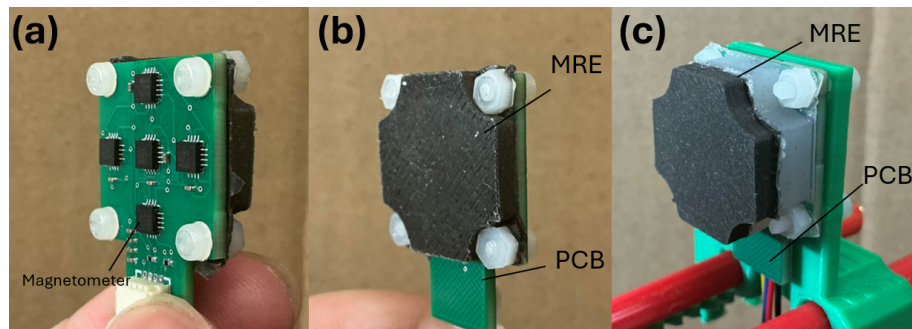


FIGURE 3.16: Manufactured a) PCB b) Prototype MRE c) Shear MRE

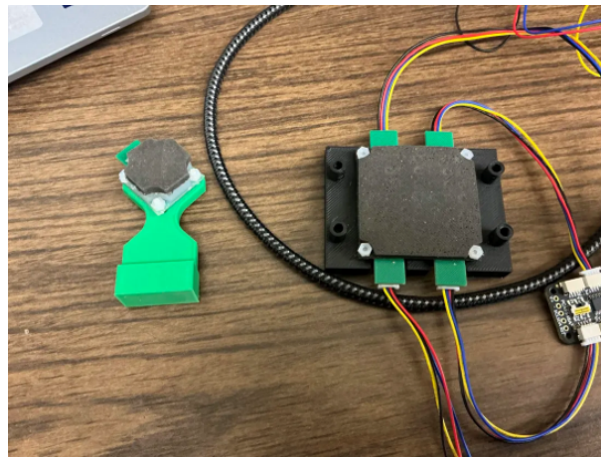


FIGURE 3.17: Final MRE designs with mounts

Chapter 4

Experiments and Results

4.1 Localization Testing

For our localization test, we expanded upon the baseplate that holds the 4 PCBs. By adding supports on the sides, we are able to create a 3D-printed grid right above the hMRE. To create indentations, we used a 3D-printed tip that will be inserted into known locations along the elastomer. To examine the affect of the tip size, we also included a varying tip sizes to get a sense of its affect. We applied a force manually at every point of interest

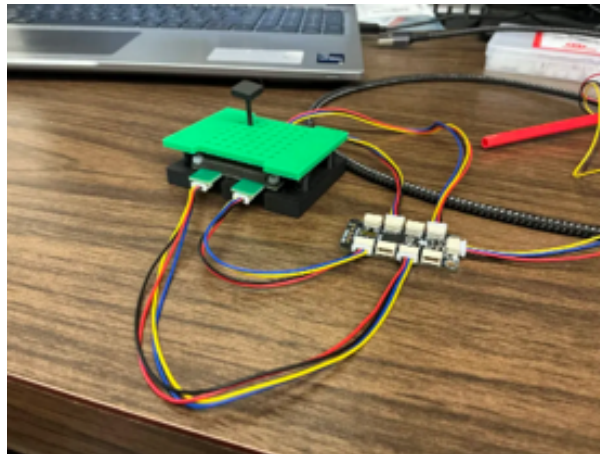


FIGURE 4.1: Localization Test (3mm End Effector)

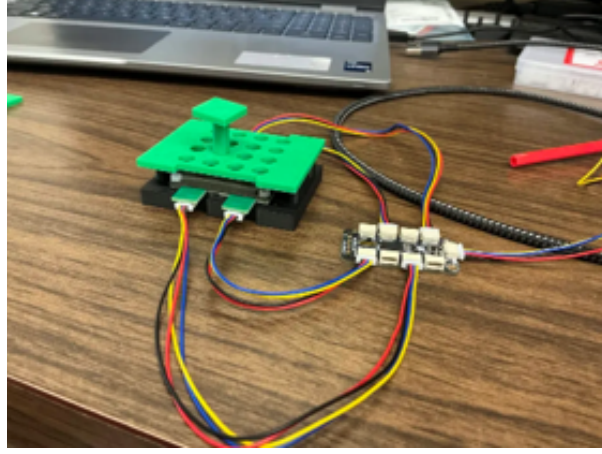


FIGURE 4.2: 7mm End Effector

To determine how accurate our localization methods are, we computed the relative error for both the weighted centroid and radial basis function methods. We conduct three trials each to get the average between the three trials. Our relative error, E , is calculated by:

$$E = \frac{\|(e_x, e_y)\|_2}{40} \cdot 100\% \quad (4.1)$$

Where (e_x, e_y) represents the absolute error for each probed position. The 40 represents the total length of our hMRE along an axis. Due to the cross-like pattern of the PCBs, the center region of our localization test is void of magnetometers.

In the plots for the relative error, the magnetometer positions are denoted by the red dots. The gaussian parameter used is 0.15 for all of these test unless noted otherwise. The probed positions are in a grid like shape, with their coordinates in the axes, and errors displayed at their position.



FIGURE 4.3: 3mm tip estimation via Weighted Centroid Relative Error



FIGURE 4.4: 3mm tip estimation via Gaussian Radial Basis Function Relative Error

In figures 4.3 and 4.4, we can see that the relative error of the probed regions near the magnetometers are low compared to other parts of the sensor, However, the error in the outer regions, increases drastically. This is caused by the lack of magnetometers that cover the entire region of interest. An example can be seen at the coordinate (5, 20), where there is no sensor to the left of this probed position, the error is at least 14%

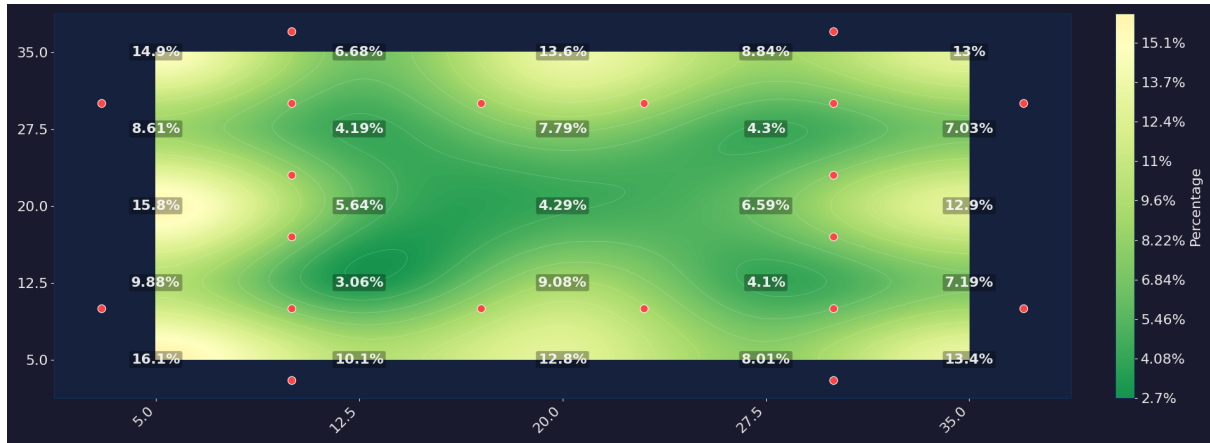


FIGURE 4.5: 5mm tip estimation via Weighted Centroid Relative Error

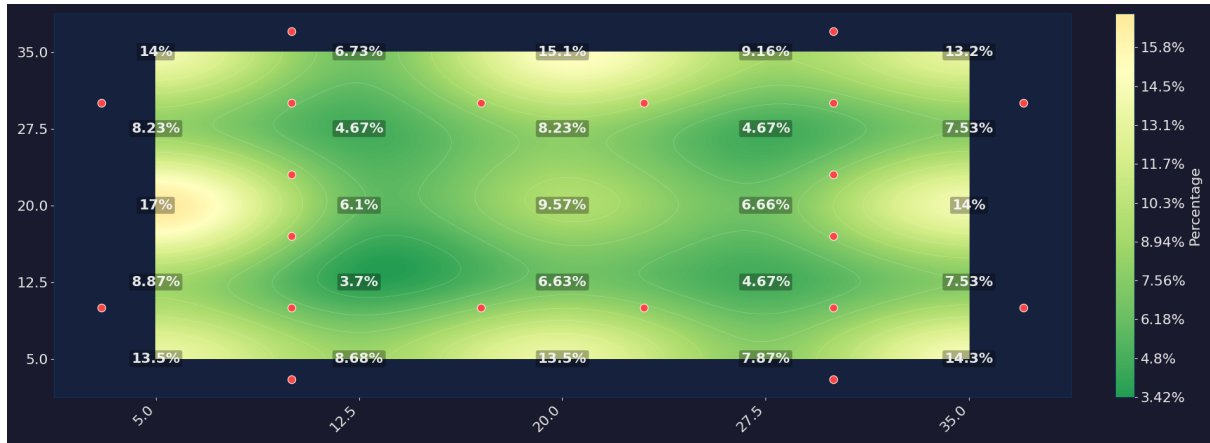


FIGURE 4.6: 5mm tip estimation via Gaussian Radial Basis Function Relative Error

For the 5mm tip test, the probed positions are no longer directly on the sensor. As a result, due to the nature of the sparse magnetometers, we can expect results with higher error. This assumption is realized as noted in figures 4.5 and 4.6. We can also note that the trend of the boundary areas with higher error is also consistent with figures 4.3 and 4.4.

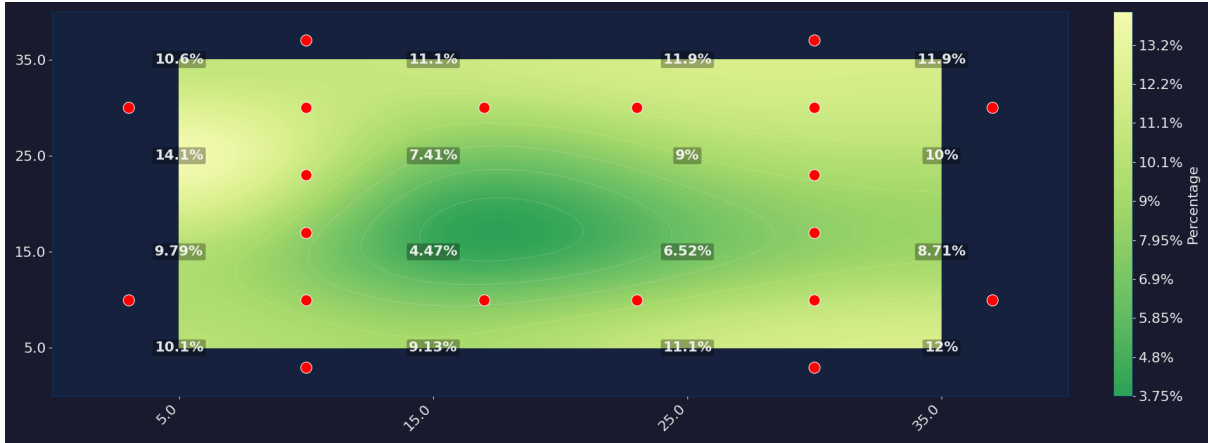


FIGURE 4.7: 7mm tip estimation via Weighted Centroid Relative Error

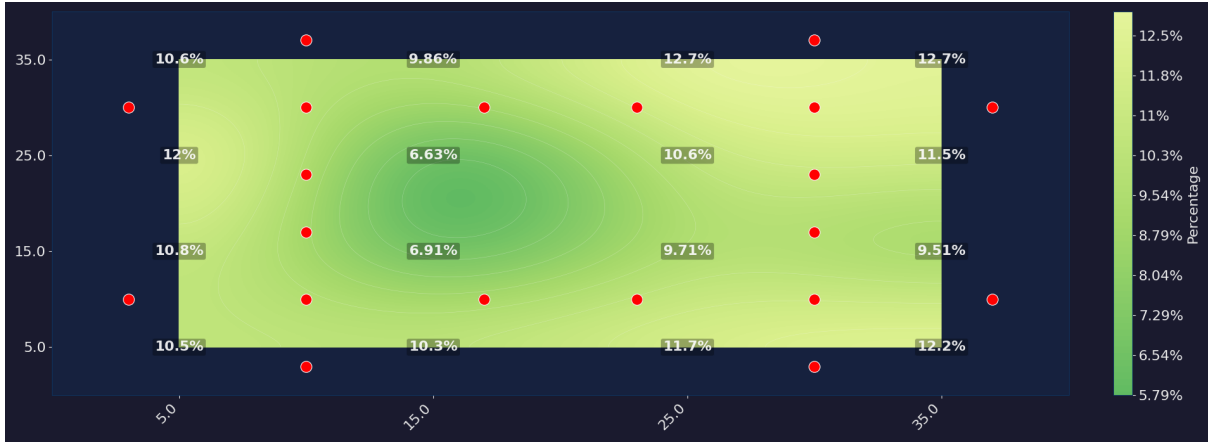


FIGURE 4.8: 7mm tip estimation via Gaussian Radial Basis Function Relative Error

In figures 4.7 and 4.8, we can continue to see the trend of the boundaries having higher errors overall. For this test, we can see that the relative error is spread out more evenly.

4.1.1 Magnetic Gradient Effects

One thing to note is that the due to how this specific hMRE was manufactured, a gradient in the magnetic field arose. In the tests so far, the gradient of the magnetic field is represented in figure 4.9. To show the impact of this gradient, we can rotate the hMRE into the formation shown in figure 4.10. These gradients were formed using our

gaussian RBF on the ambient magnetic field. Note that the smallest magnetic reading we were getting is roughly $590\mu T$, largest is $4400\mu T$, thus the span of our gradient is $3810\mu T$. This is a really large magnetic gradient.

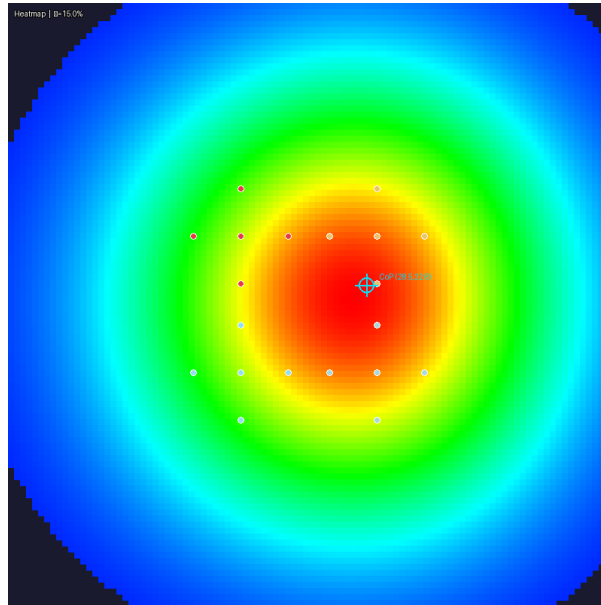


FIGURE 4.9: Gradient of Magnetic Field in Tests

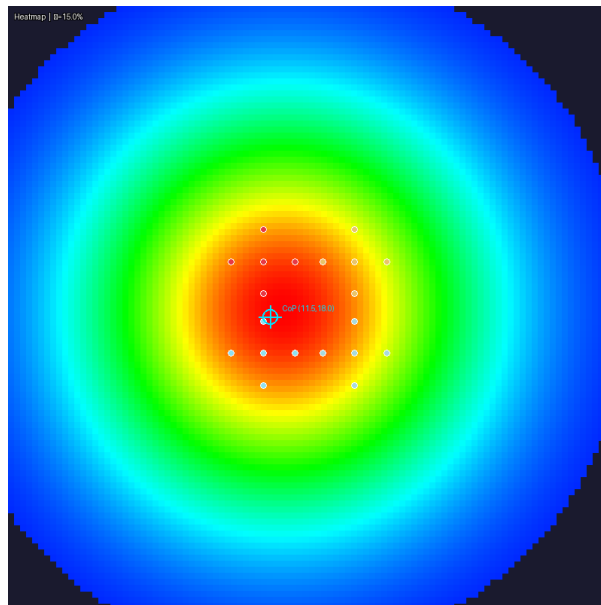


FIGURE 4.10: Gradient of Magnetic Field Rotated 180 degrees

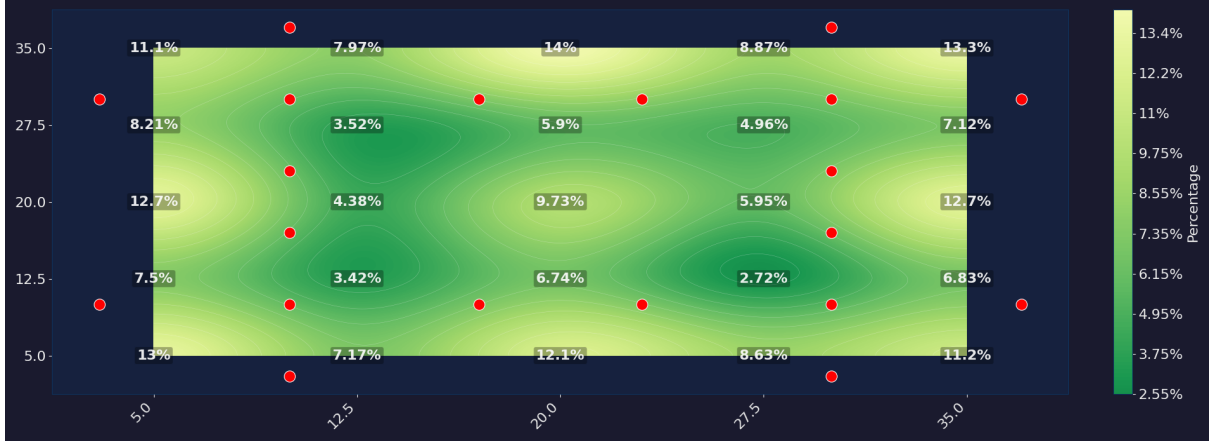


FIGURE 4.11: 5mm tip estimation via Weighted Centroid and Rotated MRE

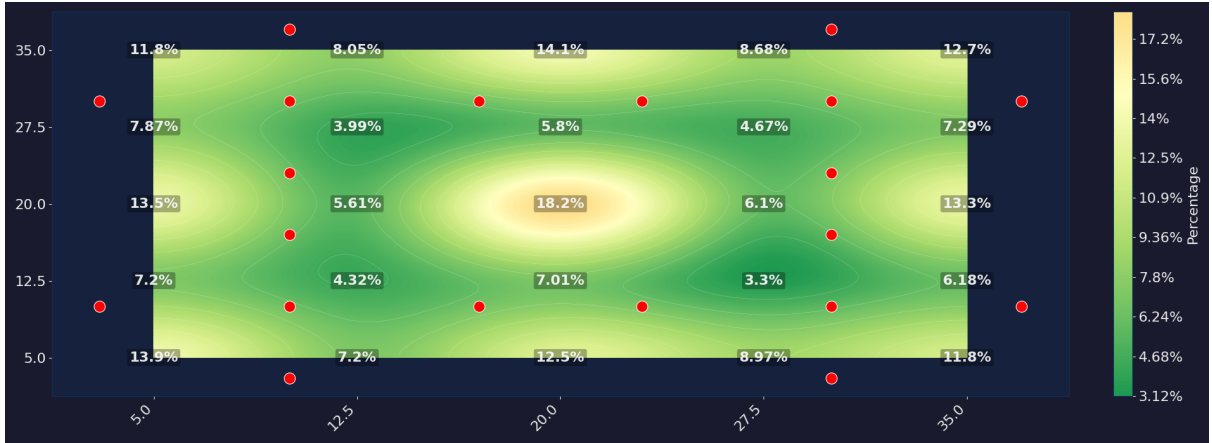


FIGURE 4.12: 5mm tip estimation via Gaussian Radial Basis Function and Rotated MRE

If we were to perform another localization test, we see that the results in figures 4.11 and 4.12 generally show a higher error. This is most likely cause due to poor alignment of the hMRE. However, to get a true estimate on how the gradient affects the readings, we must look at the absolute error for the x and y readings.

To get our absolute reading we use the following:

$$E_a = p_{reading} - p_{true} \quad (4.2)$$

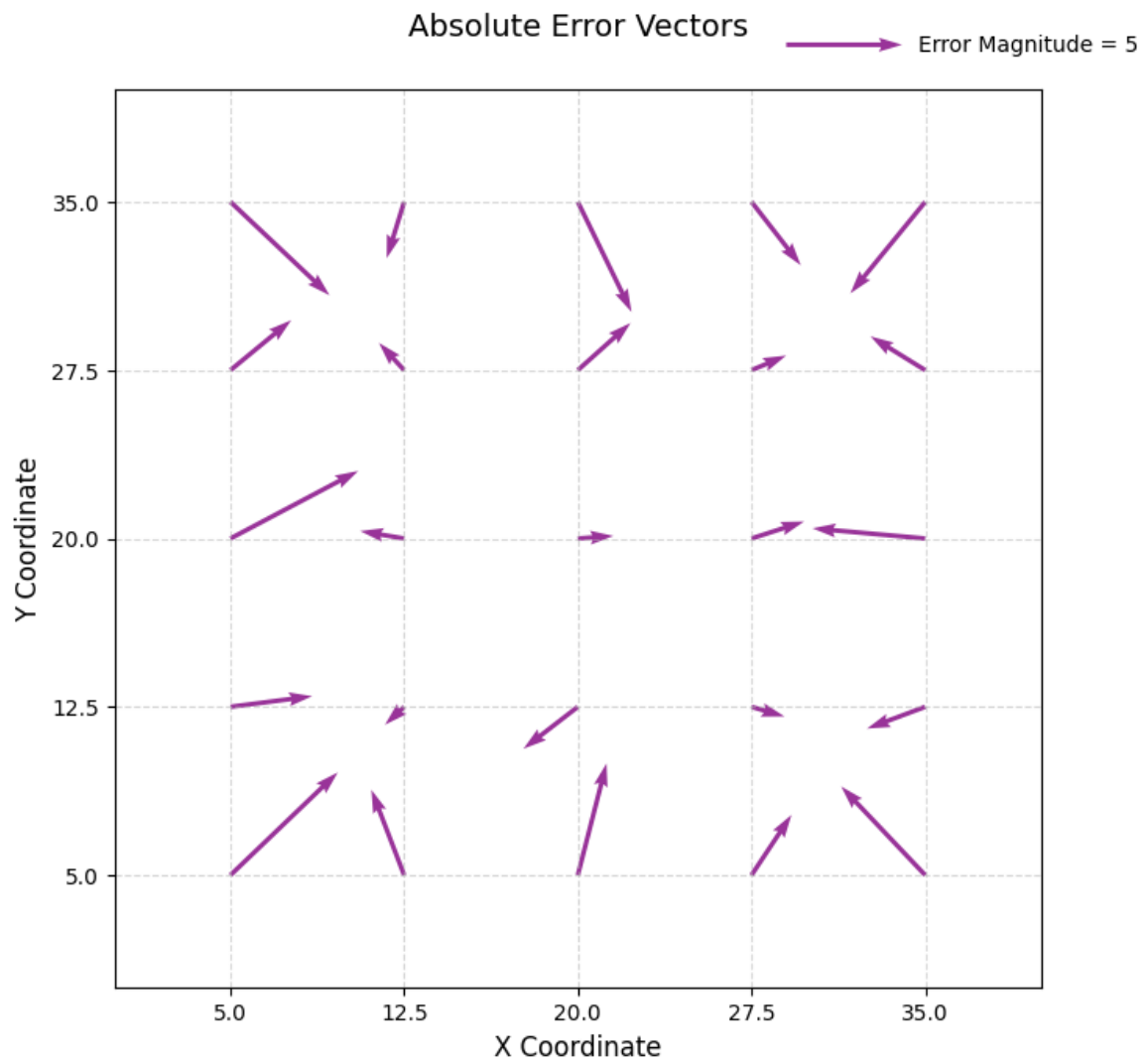


FIGURE 4.13: Un-Rotated Weighted Average Absolute Error Vector Field

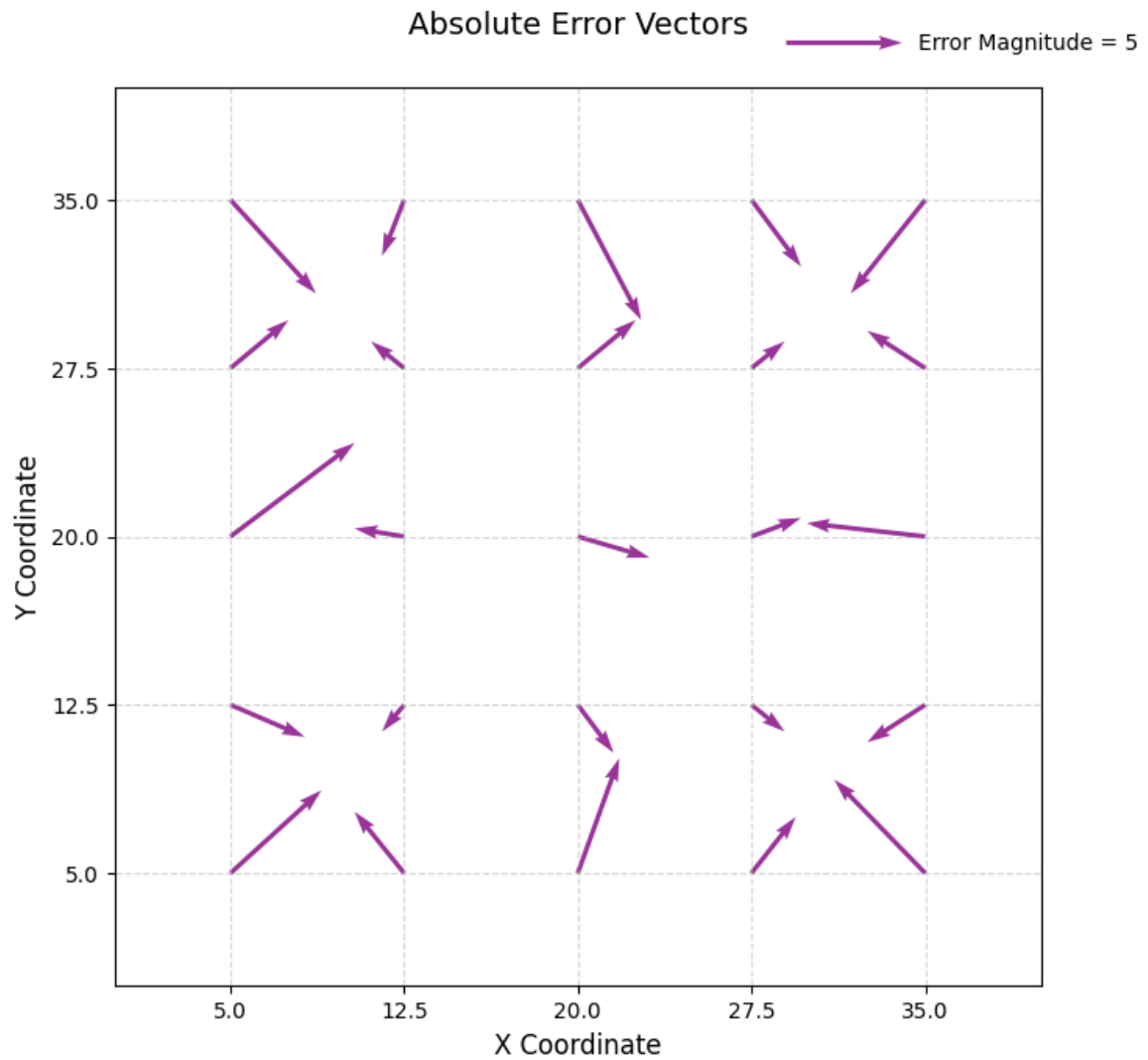


FIGURE 4.14: Un-Rotated Radial Basis Function Absolute Error Vector Field

Based on figures 4.13 and 4.14, we can see that vector fields of the absolute error generally point towards the magnetometers. However, if we look at the middle vector, the error is heavily biased toward the stronger magnetic region.

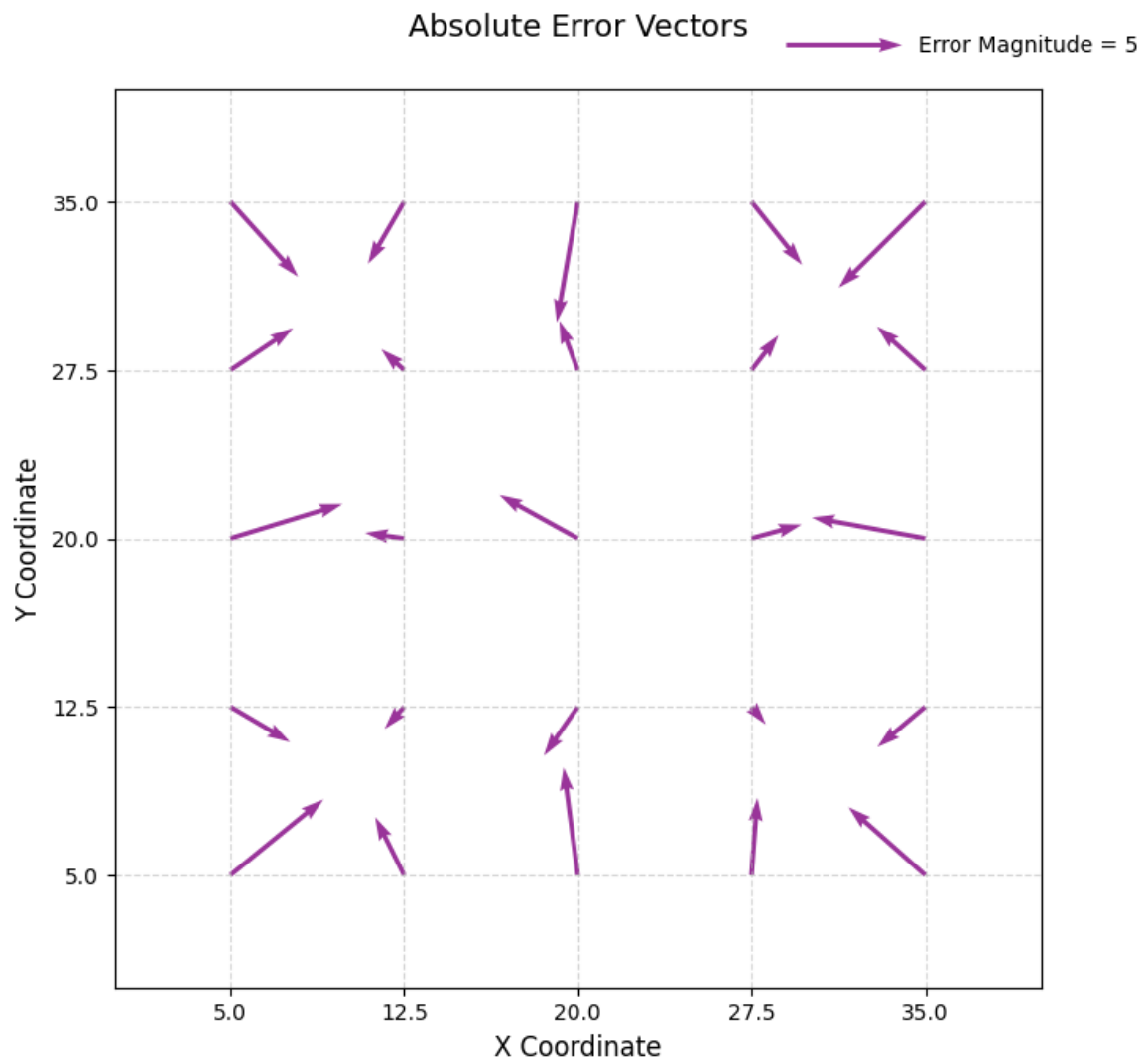


FIGURE 4.15: Rotated Weighted Average Absolute Error Vector Field

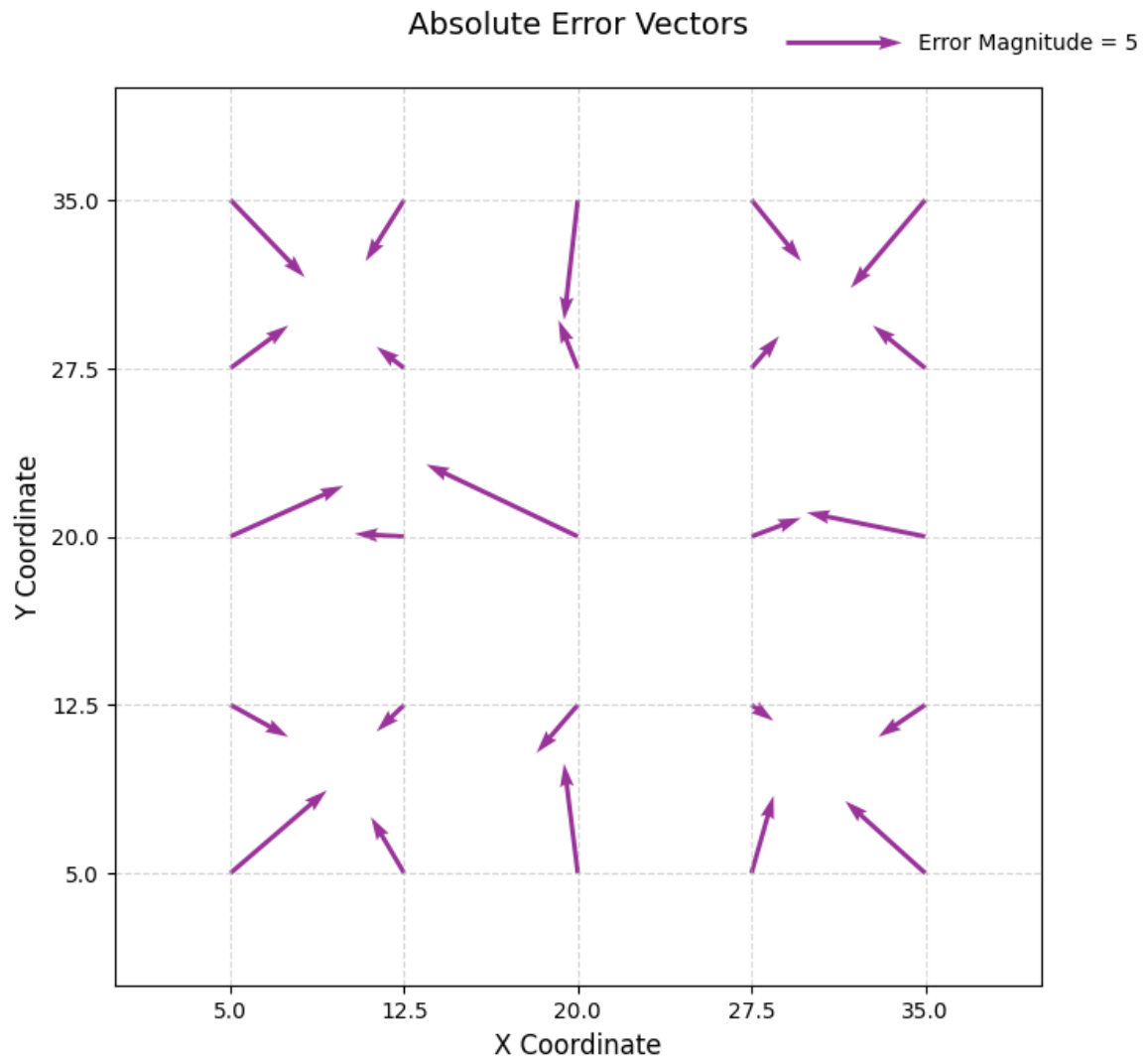


FIGURE 4.16: Rotated Radial Basis Function Absolute Error Vector Field

The trend of the error vector field pointing towards the magnetometers are consistent in the rotated hMRE figures as well 4.15 4.16. In addition, the trend of the middle vector pointing towards the stronger magnetic region remain consistent.

4.1.2 Multi Contact Estimation

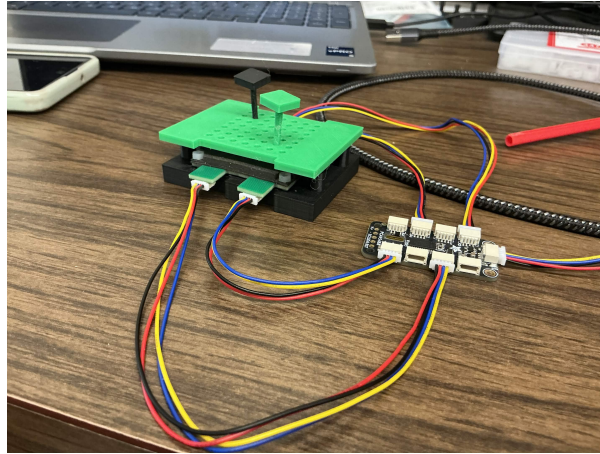


FIGURE 4.17: Multi-Contact Testing

For multi-contact estimation, we will explore how the gaussian parameter affects our sensor readings. Using the same apparatus as our localization test, we will use two tips to apply two forces in known positions. We decided to showcase the change in accuracy by pick the locations $(10, 30)$ and $(30, 10)$ on our sensor. These positions have a magnetometer directly below which removes as much error as possible besides the parameter.

We notice the with a gaussian parameter of three 4.18, each individual magnetometer is being treated as a point contact. This is because the gaussian bell curves are thin, resulting in their being 0 influence to surrounding regions.

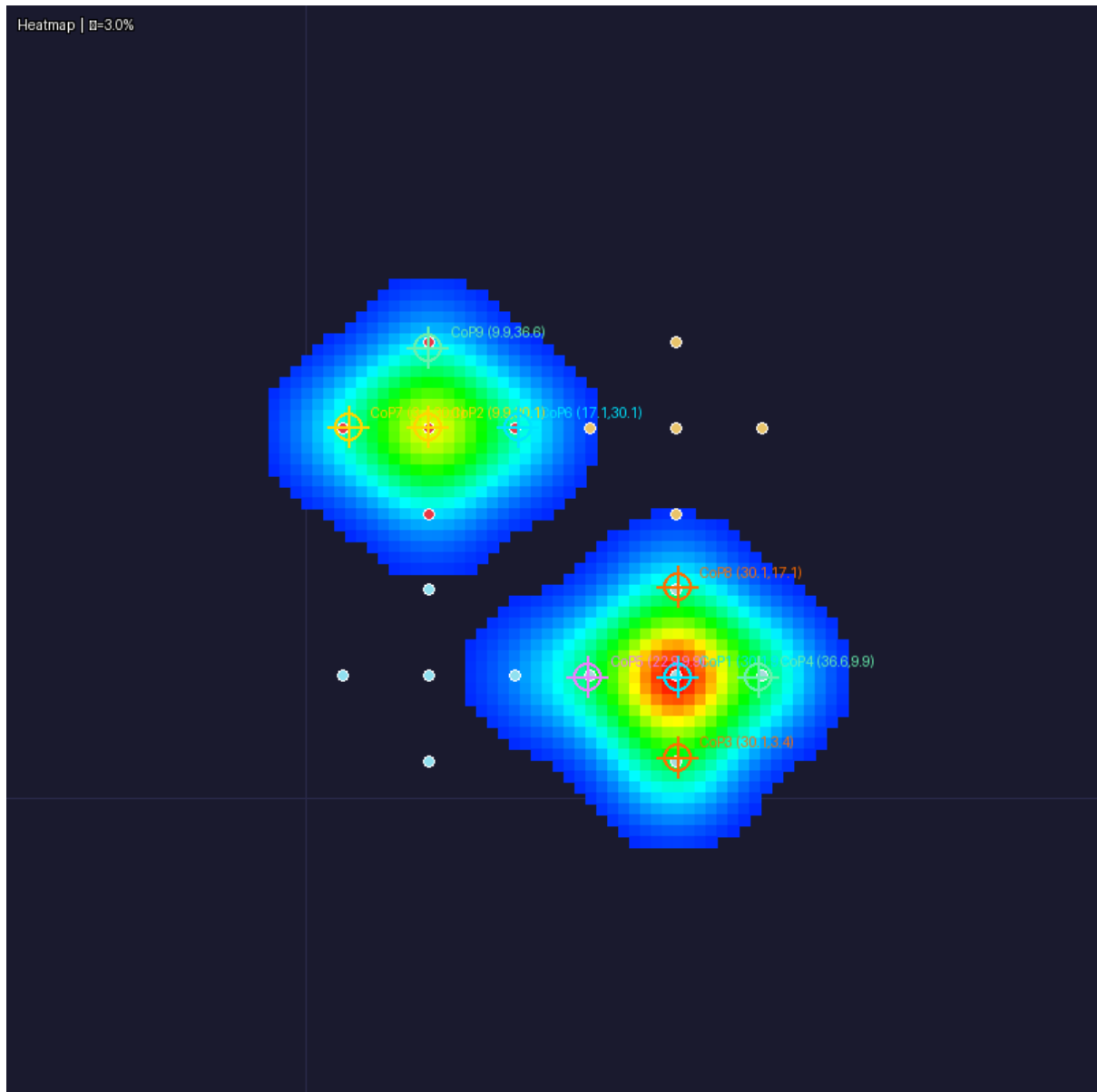


FIGURE 4.18: RBF Map with 3 Gaussian Parameter

For a gaussian parameter of 9 shown in figure 4.19, we see that our point estimations are directly where we want them to be, with minimal error.

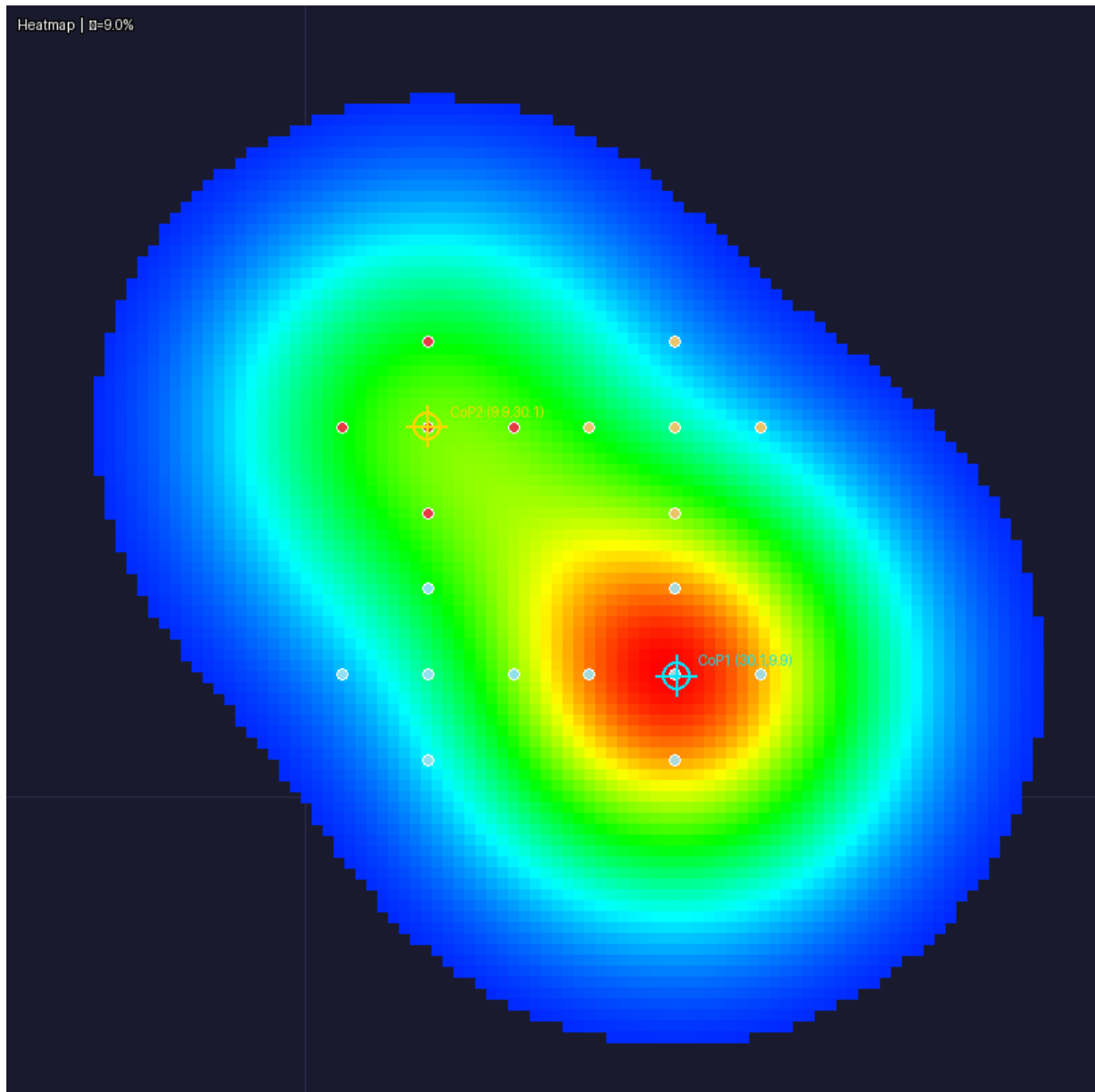


FIGURE 4.19: RBF Map with 9 Gaussian Parameter

For a gaussian parameter of 18 shown in figure 4.20, we see that the two point contacts start to merge. This is not ideal for trying to find multiple points, especially if the two point contacts are nearby.

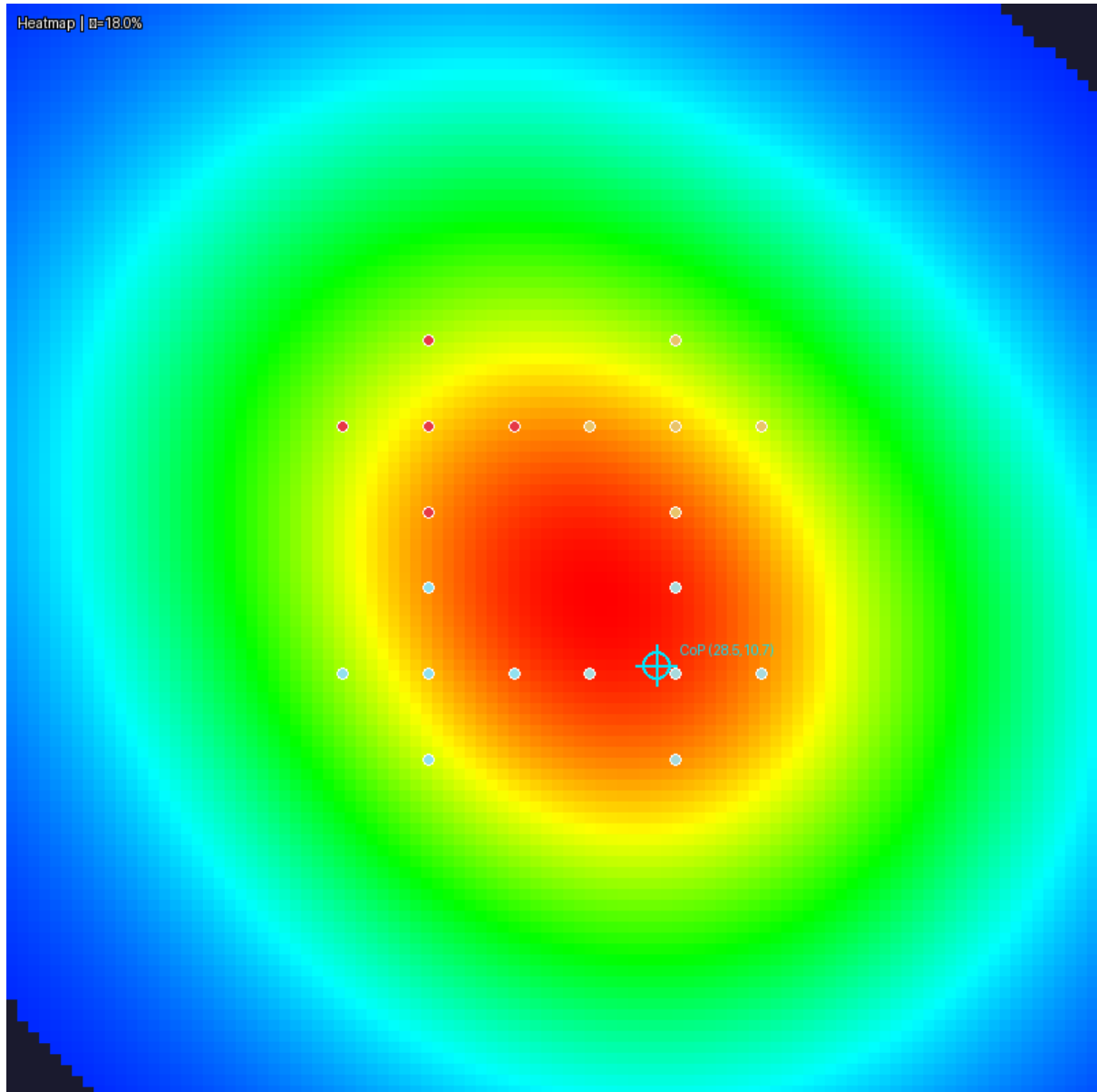


FIGURE 4.20: RBF Map with 18 Gaussian Parameter

4.2 Shear Force Testing

For our shearing test, we constructed an apparatus such that our sensor will grasp on object. To achieve the grasping operation, we decided to use clamps to hold the object in place, where one sensor is on each side.

To apply a shear force in multiple directions, instead of a complicated mechanism to apply the force around the grasped object, we instead rotated the sensor itself and

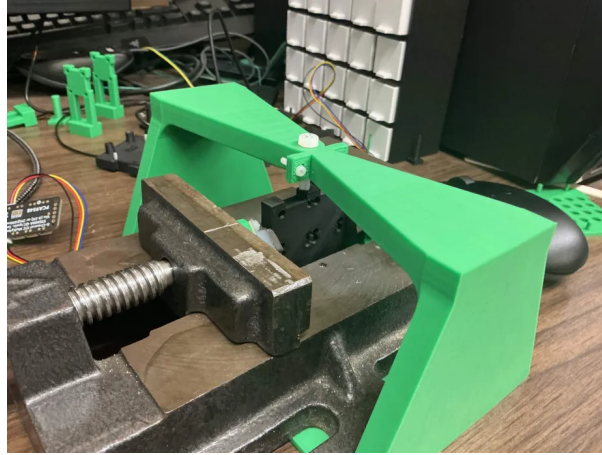


FIGURE 4.21: Shear Testing Apparatus

kept the force direction consistent. To apply the shear force in general, we will have a screw that will laterally displace the grasped object, hoping to displace the hMRE section of our tested sensor.

To keep the data similar to what a robot gripper would experience, the sensor's orientation is similar to that of how a parallel gripper would read data. The 0 degree orientation is shown in figure 3.11 and the rotation we assume is clockwise. Based on the data in figure 4.22, the shear direction estimation is accurate. This is because in a normal parallel gripper, the vices face towards each other. To rotate the vices into the same frame orientation, a mirror in the y-axis is needed to make the angles of both sensors the same. If we were to conduct this mirroring, we can validate the data collected in figure 4.23.

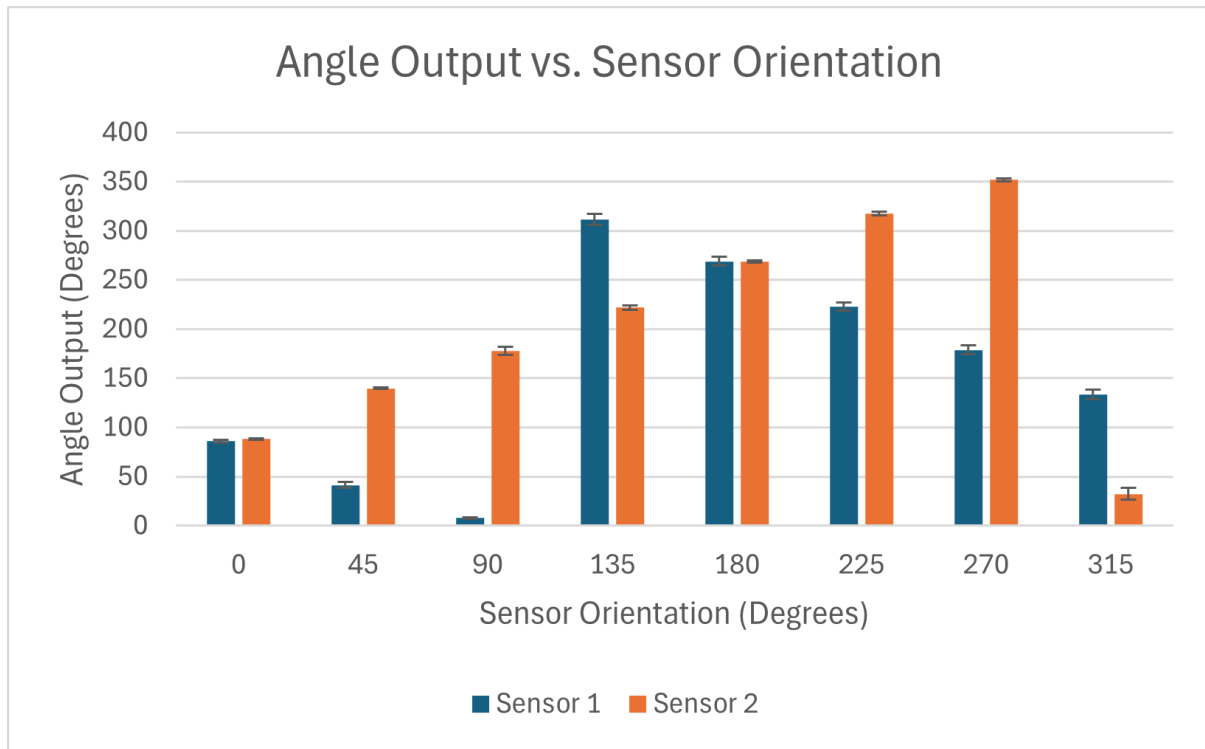


FIGURE 4.22: Shear Direction Estimation

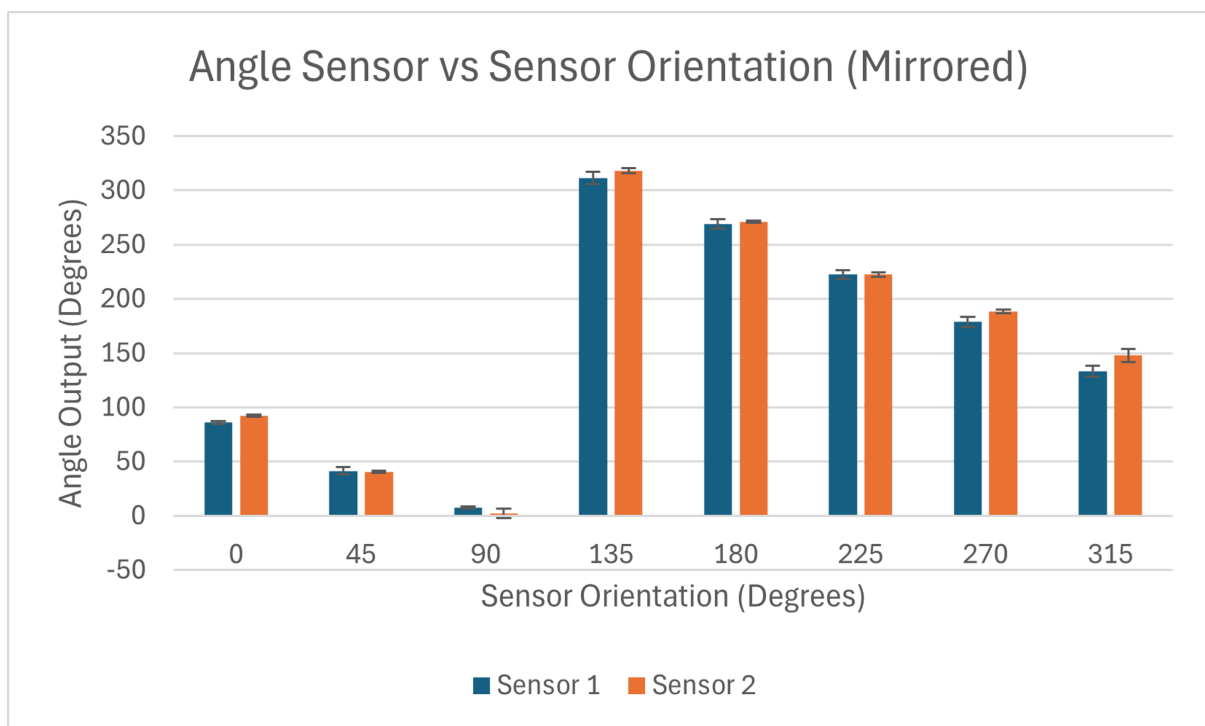


FIGURE 4.23: Shear Direction Estimation (Mirrored)

For our shear magnitude results in figures 4.24 and 4.25, we can see that these results are not ideal. This test was done on the basis that the displacement of the object would translate to sizable displacement of the hMRE. However, in the data shown in figure 4.24 and 4.25, we can see that the sensor was able to sense a displacement was occurring on the silicone. The magnitudes of the shear readings are much lower than anticipated. This drop in magnitude can be attributed to the clamp that we used. Due to it being a large, ferromagnetic object, directly against the sensor, it weakens our overall readings. The centroid generated by the ferromagnetic clamp also clamps our sensor readings toward wherever that centroid is. We even see small decreases in shear when the displacement increase. This can be attributed to slip occurring on the sensor-object contact.

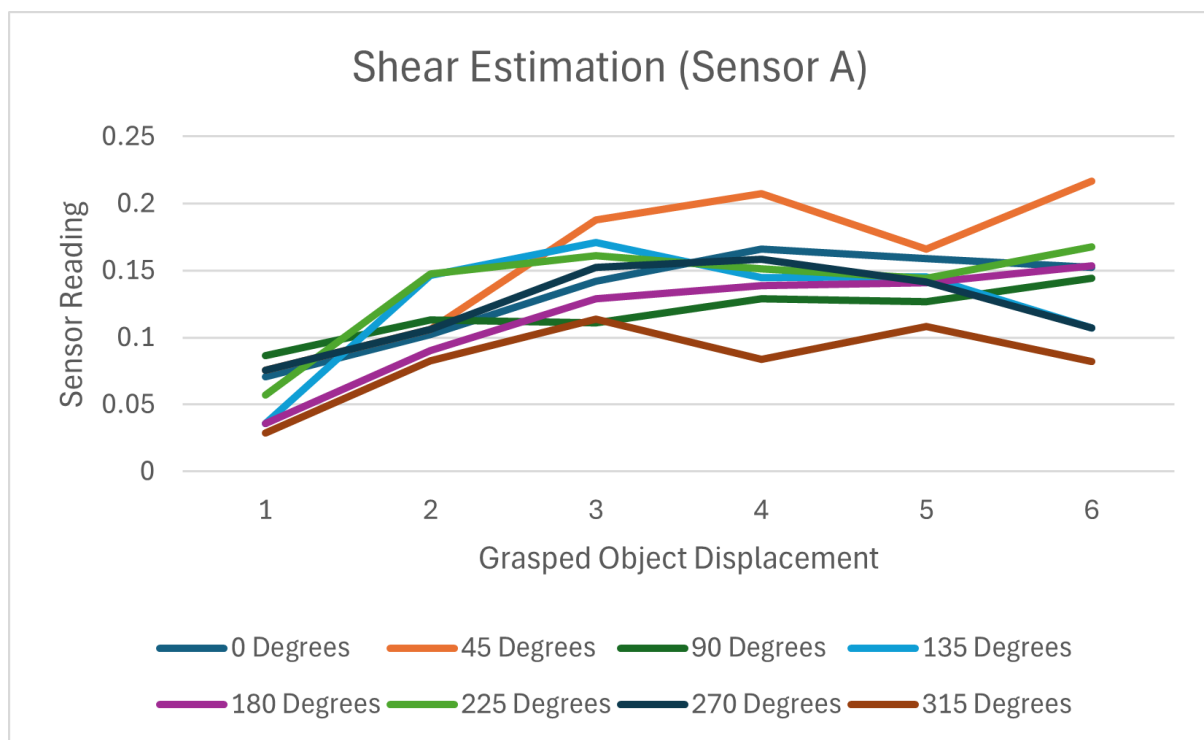


FIGURE 4.24: Shear Magnitude Readings from Sensor A

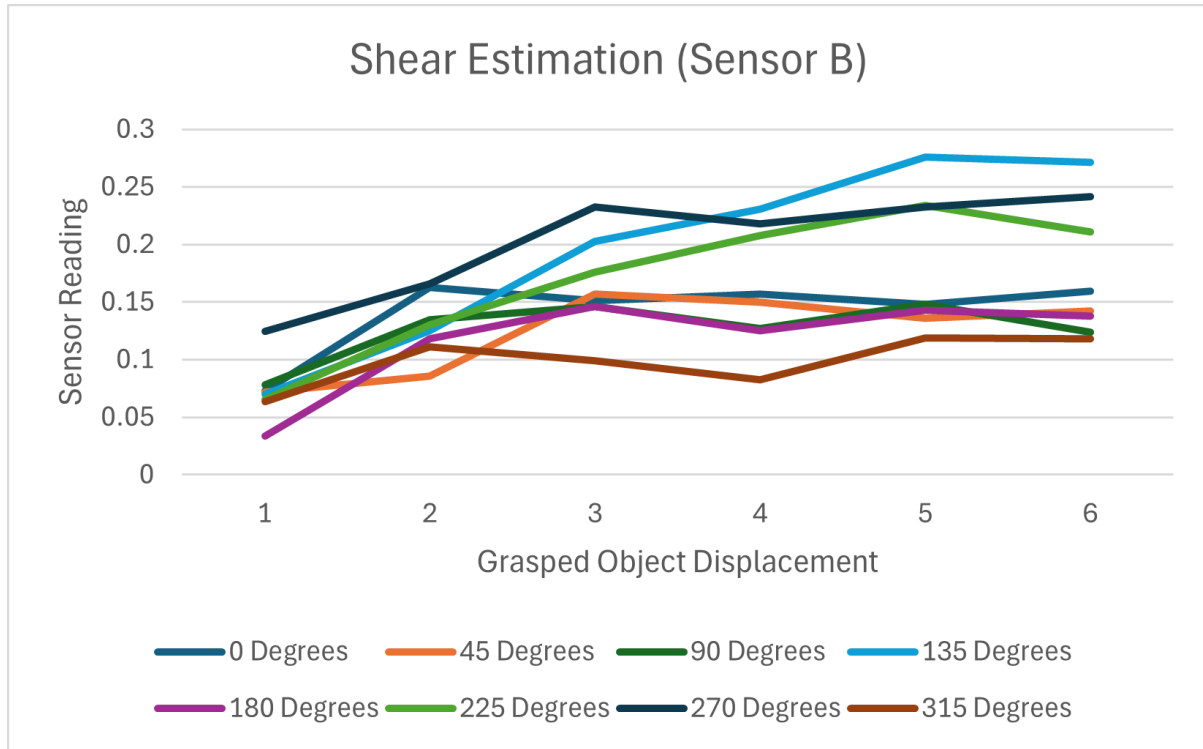


FIGURE 4.25: Shear Magnitude Readings from Sensor B

4.3 Normal Force Testing

To conduct a distributed normal force test, we must be able to have a known force push against our sensor laterally. We decided to use a simple water cup and then lay it on top of our sensor and read the data. The water cup is then measured via a scale to get the total mass, which is then converted into force. The sensor design we used is the same as the localization tests. We decided to go with this one instead of a single PCB design was the water cup was not stable on the smaller footprint that is the single PCB. If we had magnetometers in the central region, we could get more confident normal force readings.

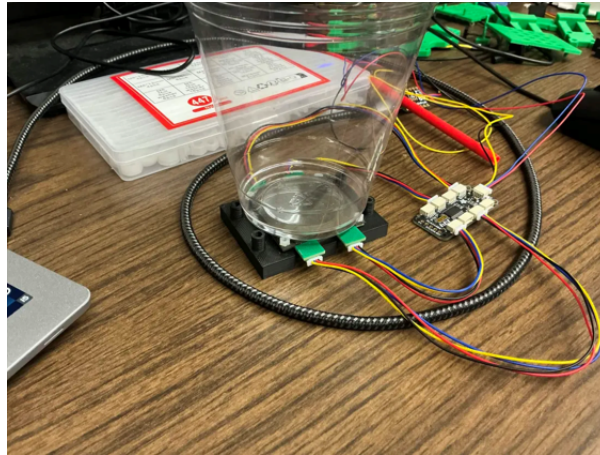


FIGURE 4.26: Normal Testing

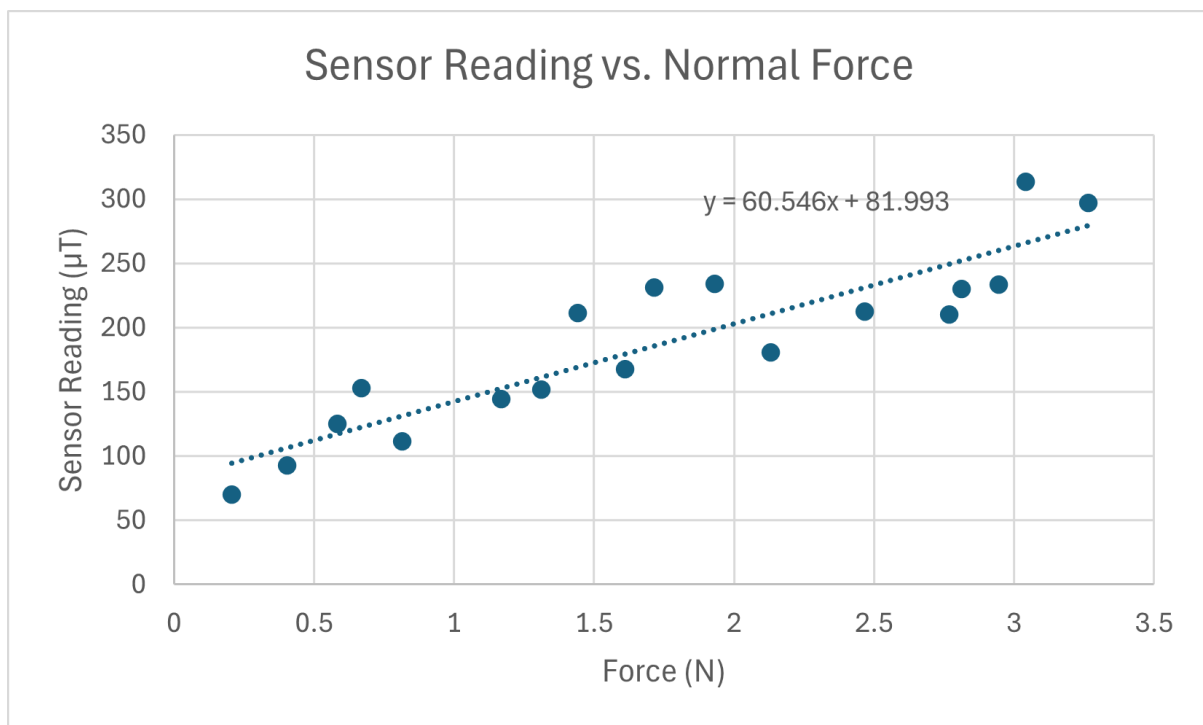


FIGURE 4.27: Sensor Readings vs. Normal Force

In our testing, we examined a linear relationship between the force our sensor output. The y-intercept encompasses the average magnetometer noise accumulated during each testing cycle. The reason why this is the case is because of the amount of magnetometers present. They each have a small amount of noise and they are compounded onto each other despite each one only supplying minimal noise.

Looking into our linear relationship results, this result makes sense, considering the range of forces we tested in. Despite expecting a non-linear relationship as found in Hossien's finite element analysis [20], they were testing with forces up to 20N. For their results for forces up to 3N, they also ended up with a linear relationship.

Chapter 5

Conclusion and Future Work

In this work, a novel soft magnetic tactile sensor was developed to provide high-fidelity, tri-axial force sensing and precise contact localization for robotic manipulation. Utilizing a hard magnetorheological elastomer embedded with melt-spun NdFeB powder positioned over a magnetometer array, the system effectively acts as a continuous, deformable permanent magnet. As physical forces are applied, localized micro-deformations translate into continuous magnetic flux fluctuations. By continuously capturing these changes, the proposed computational framework robustly maps the magnetic field disruptions into actionable measurements of normal force, shear force, and spatial contact points. This equips robotic systems with the high-fidelity tactile feedback necessary to handle delicate, contact-rich interactions and unstructured environments.

A key computational advancement of this work is the implementation of a highly accurate multi-point contact localization algorithm. While single-point estimations were initially tested using a weighted centroid method, the system leverages a spatial mapping approach using a Gaussian Radial Basis Function (RBF) to reconstruct the topological profile of the elastomer's deformation. This allows the sensor to accurately pinpoint the exact epicenter of applied forces and inherently detect multiple simultaneous contact points. Furthermore, the study effectively decouples complex multi-axial forces by mathematically tracking the continuous lateral shift of the absolute magnetic centroid for linear shear force estimation. To accurately measure applied

normal forces, the system aggregates the localized vertical magnetic field variations and passes them through an empirically derived non-linear calibration function, accounting for the hyperelastic stiffening of the elastomer.

Beyond the computational algorithms, this research contributes an improved, cost-effective manufacturing methodology for hMRE sensing applications. By refining the magnetization process and avoiding the need for an expensive pulse magnetizer, the sensor yields a strong, uniform magnetic field essential for stable and consistent performance. Ultimately, the integration of this durable, compliant sensing hardware with the continuous mapping algorithms bridges the gap in robotic perception, enabling real-time tactile awareness, slip detection, and safe manipulation of fragile objects.

While the proposed hMRE tactile sensor system offers robust tri-axial force sensing, it is currently constrained by several physical and computational limitations. Because the system relies on tracking minute fluctuations in magnetic flux, it is highly susceptible to changing external magnetic interference, such as electromagnetic noise from robotic motors or nearby ferromagnetic objects, which can induce false readings during sensing sequences. Additionally, the viscoelastic nature of the soft elastomer introduces hysteresis and creep; under dynamic or prolonged gripping, the material's failure to immediately return to its original shape can cause a baseline drift in the sensor's readings over time. Additionally, our hMRE is not perfectly elastic, meaning it doesn't even return to its original position perfectly. Finally, although the computational model effectively tracks linear shear forces, it struggles to accurately decouple highly complex, simultaneous multi-axial loading—such as combined heavy normal force, lateral shear, and severe torsion—because the resulting overlapping deformation profiles make the magnetic centroid shifts non-linear and mathematically difficult to isolate.

Some future work is to explore the increase in accuracy of the method using a pulse magnetizer for a uniform magnetic field and a more uniform grid of magnetometers

for perfect sensor resolution. Some other possible additions are to look into alternatives for finding the normal force, as it's a very noisy, non-linear system, exploring different kernels for the radial basis function (i.e., Inverse Distance Weighting (IDW) kernel), and applying the localization function onto a more complex plane.

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Appendix A

Appendix

A.1 Bill of Materials

Item	Description	Manufacturer	Purchase Link
NdFeB Powder	25 μ m Melt-Spun	Nano Chemazone	https://www.nanochemazone.com
Silicone	10A Very-Slow	Smooth-On	https://shop.smooth-on.com
PCB	Assembled	JLCPCB	https://jlcpcb.com

TABLE A.1: Bill of Materials

A.2 GitHub

Github: <https://github.com/mar-tinchan/MRE-Sensor.git>